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# Artificial intelligence and stock price crash risk: Evidence from China: A pre-registered study

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## ABSTRACT

This pre-registered study executes the empirical design approved in the associated pre-registered report (Bai and Zhao, 2025) to investigate the impact of artificial intelligence (AI) investment on stock price crash risk. Using China's "New-generation Artificial Intelligence Polit Zone Policy" as a quasi-natural experiment, we find robust evidence that AI investment significantly increases firms' stock price crash risk, mainly due to reduced information transparency and heightened managerial optimism. The effect is more pronounced for firms with lower levels of information transparency and tighter resource constraints. Furthermore, we also find the policy boosts firm value, suggesting that market optimism may drive short-term valuation gains at the cost of long-term stability. Overall, our findings highlight the unintended downside risks associated with AI investment, emphasizing the importance of transparency and governance in mitigating potential adverse outcomes.

## 1. Introduction

The rapid development of artificial intelligence (AI) has profoundly reshaped the global economy, positioning AI as a transformative general-purpose technology with far-reaching implications for firms, industries, and markets. Governments worldwide actively promote AI adoption, with China's "New-generation Artificial Intelligence Pilot Zone" policy serving as a major initiative to foster AI-driven growth. Against this backdrop, assessing the economic consequences of corporate AI investment has become both urgent and policy-relevant. Existing research has mainly focused on the real-side effects of AI, showing that adoption enhances productivity, drives innovation, and reshapes labor markets through both job displacement and creation (Cockburn et al., 2019; Acemoglu et al., 2022; Yang, 2022; Damioli et al., 2023; Babina et al., 2024). However, less is known about its capital market consequences, especially how AI investment influences investor risks and stock price stability.

Theoretically, the impact of AI investment on stock price crash risk is ambiguous. On the positive side, AI can improve information processing, enhance efficiency, and create credible growth prospects, which may reduce managers' incentives to conceal bad news and thus lower crash risk. On the negative side, the uncertainty and complexity of AI projects may amplify managerial optimism and information asymmetry, encouraging managers to withhold unfavorable information and increasing the likelihood of crashes when

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such information is eventually disclosed. To address these questions, this paper employs China's "New-generation Artificial Intelligence Pilot Zone Policy" as a quasi-natural experiment to investigate the following hypotheses:

**H1a.** AI investment will mitigate firms' stock price crash risk.

**H1b.** AI investment will increase firms' stock price crash risk.

We focus on the relationship between corporate AI investment and stock price crash risk in China for two reasons. First, China offers a nice institutional setting for a quasi-natural experiment. According to the *Stanford AI Index Report* (2025), China ranks among the global leaders in AI investment and patents. The implementation of the "New-generation Artificial Intelligence Pilot Zone Policy" represents a significant institutional initiative aimed at accelerating AI development. Its staggered rollout across regions and over time introduces exogenous variation in firms' AI exposure, enabling a more credible identification of the causal effects of AI investment on firm-level outcomes such as stock price crash risk and firm valuation. Second, the information environment and capital market are relatively weak in China compared with those of developed countries (Piotroski and Wong, 2012), which may amplify managerial tendencies to withhold bad news, making stock price crash risk more pronounced. These institutional characteristics offer a sharper lens through which to examine the unintended consequences of corporate AI investment.

This study makes three key contributions. First, we extend the focus from the positive role of AI to its potential downside. While previous research has explored the positive impact of AI investment on firms' productivity (Mihet and Philippon, 2019; Cockburn et al., 2019; Yang, 2022), labor income share (Fan et al., 2025), and innovation outcomes (Rammer et al., 2022; Babina et al., 2024), our study sheds light on the dark side of AI investment by investigating its effect on stock price crash risk, which is an important indicator of financial market stability. This perspective provides a more balanced view of the economic consequences of AI adoption. Second, we enrich the understanding of underlying mechanisms by exploring whether AI investment influences crash risk through the information transparency and managerial expectation management. This mechanism analysis deepens our insight into how technological innovation interacts with corporate disclosure behavior and market reactions, contributing to the literature on both AI and capital market dynamics. Third, we offer some insights into the policy implementation. By identifying the conditions under which AI investment may amplify downside risks, we provide evidence to help regulators design more balanced and forward-looking policies—encouraging innovation while ensuring adequate safeguards to protect market integrity and investor confidence.

This paper constitutes Phase 4 of the PBFJ pre-registration publication process (Faff, 2022), executing the study outlined in our approved pre-registered report (Bai and Zhao, 2025). Please refer to Bai and Zhao (2025) pre-registered report for the full background and motivation of the study. As per Faff (2022) guidelines, the remainder of this paper is structured as follows. Section 2 primarily includes a summary of the empirical design fully detailed in the pre-registered report. Section 3 outlines the results from the empirical analysis, followed by a discussion of the results (Section 4). Section 5 provides the conclusions and policy implications.

## 2. Pre-registered report articulation

In executing this study, the empirical design outlined in the approved pre-registered report (Bai and Zhao, 2025) was followed. Please refer to the pre-registered report for full details on the hypothesis development, data, tool and model specification. In summary, we employ China's "New-generation Artificial Intelligence Pilot Zone Policy" as the quasi-natural experiment to conduct a difference-in-difference (DID) analysis to test our hypotheses.

To measure the stock price crash risk, following Hutton et al. (2009) and He et al. (2025), we measure stock price crash risk by the negative skewness of weekly stock returns (*NCSKEW*) and the down-to-up volatility ratio (*DUVOL*), both calculated on a yearly basis. To analyze the impact of AI investment and crash risk on firm value, we follow prior studies (Wong et al., 2021; Mao et al., 2024) and use the market-to-book ratio and Tobin's Q as proxies for firm value.

The DID research design requires the identification of treatment and control groups, corresponding to firms that are, or are not, exposed to the exogenous policy shock. Accordingly, our treatment group consists of firms headquartered in cities that located in AI pilot zones, while firms in non-pilot cities serve as the control group. We construct a treatment indicator, *Treat<sub>it</sub>*, which takes a value of 1 if a firm belongs to the treatment group at year *t*, and 0 otherwise. We also define the variable of *Post<sub>t</sub>* as a time indicator that equals 1 for the post-policy period and 0 for the pre-policy period.

In addition, we include the control variables in the paper, following prior research (He et al., 2025). These variables are firm size (*Size<sub>it</sub>*), financial leverage (*Lev<sub>it</sub>*), profitability (*ROA<sub>it</sub>*), proportion of fixed assets (*PFA<sub>it</sub>*), state ownership (*SOE<sub>it</sub>*), the largest shareholder's ownership proportion (*Topone<sub>it</sub>*), managers' stock holdings (*Mshare<sub>it</sub>*), CEO-chair(wo)man duality (*Duality<sub>it</sub>*), separation of ownership and control (*Separation<sub>it</sub>*), capital intensity (*Intensity<sub>it</sub>*), book-to-market ratio (*BM<sub>it</sub>*), the average weekly stock returns (*RET<sub>it</sub>*), the volatility of weekly stock returns (*Sigma<sub>it</sub>*) and share turnover (*Turnover<sub>it</sub>*). Moreover, since the dependent variable in this study reflects the crash risk in the subsequent period, we follow Wang et al. (2024) and include the current period's stock price crash risk as control variable. Definitions for these variables are detailed in Appendix. Additionally, we incorporate year, industry and city levels of fixed effects to control unobserved factors that might influence our results. Standard errors are clustered at the firm level. To test our hypotheses, we formulate the following DID model, as specified in Eq. (1).

$$CrashRisk_{i,t+1} = \alpha_0 + \alpha_1 Treat_{i,t} \times Post_{i,t} + \alpha_2 Controls_{i,t} + Industry + Year + City + \varepsilon_{i,t} \quad (1)$$

**Table 1**  
Univariate statistics.

| Panel A: Descriptive statistics |        |        |           |        |        |
|---------------------------------|--------|--------|-----------|--------|--------|
| Variable                        | Obs    | Mean   | Std. Dev. | Min    | Max    |
| <i>NCSKEW</i>                   | 25,746 | −0.430 | 0.747     | −2.623 | 1.617  |
| <i>DUVOL</i>                    | 25,746 | −0.299 | 0.479     | −1.448 | 0.937  |
| <i>Treat</i>                    | 25,746 | 0.419  | 0.493     | 0      | 1      |
| <i>Size</i>                     | 25,746 | 22.544 | 1.348     | 19.475 | 27.285 |
| <i>Lev</i>                      | 25,746 | 0.461  | 0.202     | 0.057  | 0.988  |
| <i>ROA</i>                      | 25,746 | 0.032  | 0.064     | −0.342 | 0.199  |
| <i>PFA</i>                      | 25,746 | 0.224  | 0.165     | 0.002  | 0.706  |
| <i>SOE</i>                      | 25,746 | 0.449  | 0.497     | 0      | 1      |
| <i>Growth</i>                   | 25,746 | 0.113  | 0.316     | −0.651 | 1.689  |
| <i>Top1</i>                     | 25,746 | 0.353  | 0.150     | 0.081  | 0.739  |
| <i>Mshare</i>                   | 25,746 | 3.409  | 9.290     | 0      | 49.34  |
| <i>Duality</i>                  | 25,746 | 0.231  | 0.421     | 0      | 1      |
| <i>Separation</i>               | 25,746 | 0.049  | 0.075     | 0      | 0.284  |
| <i>Intensity</i>                | 25,746 | 2.691  | 3.389     | 0.397  | 42.449 |
| <i>BM</i>                       | 25,746 | 0.662  | 0.258     | 0.112  | 1.195  |
| <i>RET</i>                      | 25,746 | 0.002  | 0.008     | −0.019 | 0.036  |
| <i>Sigma</i>                    | 25,746 | 0.059  | 0.023     | 0.024  | 0.144  |
| <i>Turnover</i>                 | 25,746 | −0.059 | 0.387     | −2.068 | 1.071  |

| Panel B: Correlation matrix |               |              |             |            |            |            |
|-----------------------------|---------------|--------------|-------------|------------|------------|------------|
|                             | <i>NCSKEW</i> | <i>DUVOL</i> | <i>Size</i> | <i>Lev</i> | <i>ROA</i> | <i>SOE</i> |
| <i>NCSKEW</i>               | 1             | 0.878***     | 0.020***    | −0.038***  | 0.085***   | −0.061***  |
| <i>DUVOL</i>                | 0.872***      | 1            | 0.009       | −0.043***  | 0.073***   | −0.061***  |
| <i>Size</i>                 | 0.013**       | 0.008        | 1           | 0.458***   | 0.005      | 0.299***   |
| <i>Lev</i>                  | −0.040***     | −0.044***    | 0.441***    | 1          | −0.416***  | 0.228***   |
| <i>ROA</i>                  | 0.052***      | 0.045***     | 0.062***    | −0.363***  | 1          | −0.117***  |
| <i>SOE</i>                  | −0.064***     | −0.064***    | 0.305***    | 0.226***   | −0.057***  | 1          |

Panel A of Table 1 reports the descriptive statistics of all variables used in Model (1). All the continuous variables are winsorized at the 1 and 99 percentage points, respectively. The sample period ranges from 2011 to 2024. Observations that have missing values in any of the regressors are excluded from the samples used for the multivariate tests. Panel B of Table 1 provides the Spearman and Pearson correlation coefficients for the main variables involved in the baseline regression. All the continuous variables are winsorized at the 1 and 99 percentage points, respectively. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

### 3. Empirical analysis

#### 3.1. Descriptive statistics

Panel A of Table 1 reports the descriptive statistics for the variables used in Eq. (1). During the sample period, the mean values of *NCSKEW* and *DUVOL* are −0.430 and −0.299, with a standard deviation of 0.747 and 0.479. The mean value of *Treat* is 0.419, indicating that approximately 41.9 % of our sample firms are exposed to the AI pilot zone policy and thus belong to the treatment group. Panel B of Table 1 shows the Spearman and Pearson correlation matrix of main variables. *NCSKEW* and *DUVOL* are highly correlated, with statistically significant coefficients of 0.872 and 0.878, suggesting that these two variables capture the underlying same construct for stock price crash risk. Correlation coefficients among other variables are below 0.5, indicating that multicollinearity is of less concern in our regression analyses.

#### 3.2. Main confirmatory analysis (pre-registered)

This section provides the results from the empirical design approved in the pre-registered report (Bai and Zhao, 2025), including baseline results, robustness tests, mechanism analysis and additional tests.

##### 3.2.1. Baseline results

Table 2 reports the estimation results based on Eq. (1). Column (1) and Column (4) report regressions including only the key explanatory variable (*Treat*×*Post*) along with industry, year, and city fixed effects. Column (2) and Column (5) extend the model by incorporating a set of control variables, while Column (3) and Column (6) include the full set of controls. The coefficients on *Treat*×*Post* are positive and statistically significant across all specifications for both *NCSKEW* and *DUVOL*. These results provide support for H1, which states that firms exposed to the AI pilot zone policy experience a higher stock price crash risk relative to unaffected firms. Moreover, the estimated effects of the control variables are consistent with prior studies (e.g., Piotroski et al., 2015; Wang et al., 2024; He et al., 2025).

**Table 2**  
Baseline regression.

|                            | (1)                   | (2)                  | (3)                   | (4)                   | (5)                  | (6)                   |
|----------------------------|-----------------------|----------------------|-----------------------|-----------------------|----------------------|-----------------------|
|                            | $NCSKEW_{t+1}$        | $NCSKEW_{t+1}$       | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         | $DUVOL_{t+1}$        | $DUVOL_{t+1}$         |
| <i>Treat</i> × <i>Post</i> | 0.043**<br>(2.28)     | 0.044**<br>(2.36)    | 0.043**<br>(2.34)     | 0.032***<br>(2.66)    | 0.033***<br>(2.73)   | 0.030***<br>(2.61)    |
| $NCSKEW_t$                 |                       |                      | 0.042***<br>(5.98)    |                       |                      |                       |
| $DUVOL_t$                  |                       |                      |                       |                       |                      | 0.042***<br>(6.17)    |
| $Size_t$                   |                       | 0.013**<br>(2.44)    | 0.072***<br>(12.59)   |                       | 0.003<br>(0.92)      | 0.036***<br>(9.58)    |
| $Lev_t$                    |                       | −0.022<br>(−0.64)    | −0.056<br>(−1.62)     |                       | −0.025<br>(−1.12)    | −0.038*<br>(−1.73)    |
| $ROA_t$                    |                       | 0.384***<br>(4.11)   | −0.237**<br>(−2.39)   |                       | 0.227***<br>(3.77)   | −0.174***<br>(−2.73)  |
| $PAF_t$                    |                       |                      | 0.003<br>(0.07)       |                       |                      | −0.023<br>(−0.90)     |
| $SOE_t$                    |                       |                      | −0.057***<br>(−4.45)  |                       |                      | −0.043***<br>(−5.24)  |
| $Growth_t$                 |                       |                      | −0.005<br>(−0.29)     |                       |                      | 0.001<br>(0.03)       |
| $Top1_t$                   |                       |                      | −0.031<br>(−0.79)     |                       |                      | 0.005<br>(0.20)       |
| $Mshare_t$                 |                       |                      | 0.174***<br>(2.71)    |                       |                      | 0.093**<br>(2.23)     |
| $Duality_t$                |                       |                      | 0.016<br>(1.20)       |                       |                      | 0.009<br>(1.02)       |
| $Separation_t$             |                       |                      | −0.053<br>(−0.76)     |                       |                      | −0.017<br>(−0.37)     |
| $Intensity_t$              |                       |                      | 0.001<br>(0.18)       |                       |                      | 0.001<br>(0.63)       |
| $BM_t$                     |                       |                      | −0.440***<br>(−15.06) |                       |                      | −0.247***<br>(−13.07) |
| $RET_t$                    |                       |                      | 12.000***<br>(14.50)  |                       |                      | 8.482***<br>(15.26)   |
| $Sigma_t$                  |                       |                      | −0.918***<br>(−2.77)  |                       |                      | −1.101***<br>(−5.18)  |
| $Turnover_t$               |                       |                      | −0.089***<br>(−5.95)  |                       |                      | −0.055***<br>(−5.96)  |
| _cons                      | −0.437***<br>(−72.75) | −0.725***<br>(−6.55) | −1.665***<br>(−13.80) | −0.300***<br>(−77.79) | −0.366***<br>(−5.09) | −0.841***<br>(−10.83) |
| <i>Industry FE</i>         | Y                     | Y                    | Y                     | Y                     | Y                    | Y                     |
| <i>Year FE</i>             | Y                     | Y                    | Y                     | Y                     | Y                    | Y                     |
| <i>City FE</i>             | Y                     | Y                    | Y                     | Y                     | Y                    | Y                     |
| <i>N</i>                   | 25,746                | 25,746               | 25,746                | 25,746                | 25,746               | 25,746                |
| <i>adj. R<sup>2</sup></i>  | 0.03                  | 0.03                 | 0.06                  | 0.04                  | 0.04                 | 0.06                  |

Table 2 presents the regression based on Eq. (1). In Columns (1)–(3), the dependent variables are  $NCSKEW$ . We list the results without and with control variables, respectively. In Columns (4)–(6), the dependent variables are  $DUVOL$ . We list the results without and with control variables, respectively. The robust t-values are clustered at the firm level and presented in parentheses. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

### 3.2.2. Robustness tests

Although our DID approach helps mitigate endogeneity concerns arising from confounding effects, the results may still be affected by omitted variables, sample selection, and reserve causality biases. To further address these potential endogeneity issues, we conduct a series of additional robustness tests in this section.

**3.2.2.1. Tests of parallel trends assumption.** The validity of the DID approach relies crucially on the parallel trends assumption, which requires that the trends of stock price crash risk for the treatment group should be similar with those of the control group in the pre-event period (i.e., before the implement of AI pilot zone policy reform). To test this assumption, we formulate the following Eq. (2):

$$NCSKEW_{i,t+1} \text{ or } DUVOL_{i,t+1} = \beta_0 + \beta_1 Treat_t \times Pre\_4 + \beta_2 Treat_t \times Pre\_3 + \beta_3 Treat_t \times Pre\_2 + \beta_4 Treat_t \times Pre\_1 + \beta_5 Treat_t \times Post\_1 + \beta_6 Treat_t \times Pos\_2 + \beta_7 Treat_t \times Pos\_3 + \beta_8 Controls_{i,t} + Industry + Year + City + \varepsilon_{i,t} \quad (2)$$

where  $Pre\_4$ ,  $Pre\_3$ ,  $Pre\_2$ ,  $Pre\_1$ ,  $Post\_1$ ,  $Post\_2$ , and  $Post\_3$  are the year dummies for the 7-year periods.

Appendix A.2 reports the results of Eq. (2). The coefficients on interaction terms,  $Treat \times Pre\_4$ ,  $Treat \times Pre\_3$ ,  $Treat \times Pre\_2$ ,  $Treat \times$

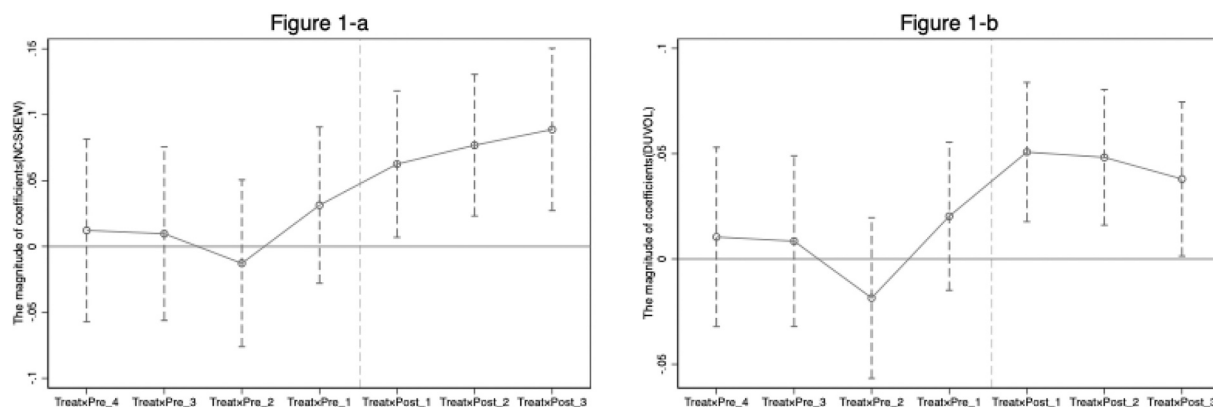


Fig. 1. Parallel trends tests.

Fig. 1 presents the results of the coefficient test of the parallel trends assumption for the DID regression estimation. Specifically, Fig. 1-a (1-b) shows the graphical diagnostic of parallel trends assumption for the DID regression where *NCSKEW* (*DUVOL*) is the dependent variable.

*Pre\_1*, are not statistically significant, indicating that there are no significant differences in the stock price crash risk trends between the treatment and control groups prior to the AI pilot zone policy. This finding supports the parallel trends assumption underlying our DID research design. The coefficients on interaction terms, *Treat*  $\times$  *Post\_1*, *Treat*  $\times$  *Post\_2*, and *Treat*  $\times$  *Post\_3*, are all significantly positive, suggesting that AI pilot zone reform increases the stock price crash risk for treated firms in each year following the policy implementation.

Fig. 1 illustrates the dynamic economic effects of the reform over different years. It reveals that the coefficients for the periods before the AI pilot zone reform are not significantly different from zero. However, the coefficients for the periods after the reform are significantly positive, suggesting that the policy effect becomes prominent after implementation. These results provide additional support for the parallel trends assumption and indicate that the increased risk of stock price crashes is attributed to the AI pilot zone reform rather than potential omitted time-series factors.

**3.2.2.2. Placebo tests.** Following previous studies (Alder et al., 2016; He et al., 2025), we conduct a series of placebo tests to examine whether our baseline regression results are driven by random factors or omitted variables. Specifically, we use *NCSKEW* and *DUVOL* as the dependent variables and design three types of placebo tests: in-time, in-space, and mixed.

First, we perform in-time placebo tests by artificially moving the AI pilot policy forward by 1–10 years and re-estimating the DID models using *NCSKEW* and *DUVOL*. As shown in Fig. 2-a and 2-b, all placebo estimates fluctuate around zero and are statistically insignificant, indicating that our results are not driven by pre-existing time trends.

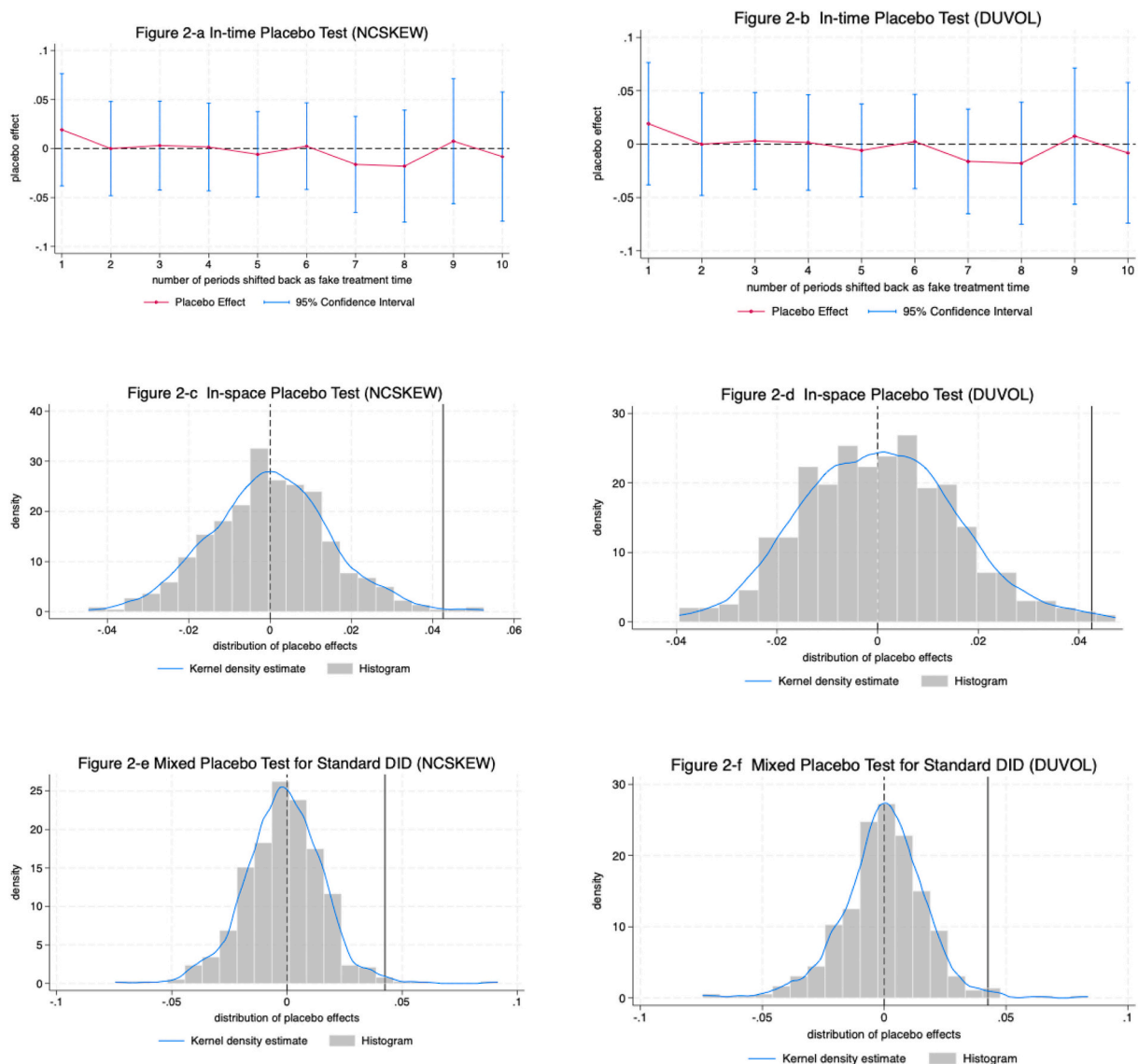
Second, we conduct in-space placebo tests by randomly assigning a subset of cities as a pseudo-treatment group and repeating the DID estimation 500 times. As shown in Fig. 2-c and 2-d, the estimated treatment effect lies in the right tail of the placebo distribution, and both two-sided and right-tailed *p*-values are statistically significant at the 1 % level, strongly rejecting the null hypothesis of no treatment effect.

Finally, we implement mix-placebo tests, in which both treatment status and treatment timing are randomly reassigned across firms and years. This procedure is repeated 500 times. Fig. 2-e and Fig. 2-f show that the actual estimated treatment effect lies in the right tail of the placebo distribution, and the corresponding *p*-values are significant at the 1 % level, confirming the robustness of our main results.

**3.2.2.3. PSM-DID.** To further address potential endogeneity arising from systematic differences between the treatment and control groups, we implement propensity score matching (PSM) and use the post-matched sample to run our DID regression. The matching procedure aims to construct a control group that closely resembles the treatment group in the AI pilot zones, based on key covariates, including firm size (*Size*), financial leverage (*Lev*), profitability (*ROA*), proportion of fixed assets (*PFA*), sales grow (*Grow*), and book-to-market ratio (*BM*). Propensity scores are estimated using a logit model, and the baseline DID model is then applied to the matched sample.

Appendix A.3 reports the PSM-DID results. Columns (1) and (2) present estimates based on the nearest neighbor matching, where each treated firm is matched to control firms with the closest propensity scores. Columns (3) and (4) show results based on radius matching, which pairs treated firms with all control firms within a specified caliper of the propensity score. Columns (5) and (6) present estimates using kernel matching, which constructs a weight average of control firms for each treated firm. Across all matching methods, the coefficients on *Treat*  $\times$  *Post* remain positive and statistically significant, further confirming the robustness of our findings.

**3.2.2.4. Alternative measures of stock price crash risk.** Following previous research (Hutton et al., 2009; He et al., 2025), we use two alternative proxies for stock price crash risk, *CRASH1* and *CRASH2*, to re-test our main hypothesis. *CRASH1* is a dummy variable that equals to 1 if a firm experiences at least one crash week during the fiscal year, and 0 otherwise. *CRASH2* is a continuous measure,



**Fig. 2.** Placebo tests.

Fig. 2 presents the distribution of placebo estimates using both NCSKEW and DUVOL as dependent variables under three types of placebo tests: in-time placebo test, in-space placebo test, and mixed placebo test. In the time placebo tests (Fig. 2-a and 2-b), we randomly assign pseudo-treatment years and estimate the treatment effects by artificially postponing the policy shock by 1 to 10 years. In the space placebo tests (Fig. 2-c and 2-e), we randomly assign a pseudo-treatment group of cities and repeat 500 times to generate the distribution of placebo effects. In the mixed placebo tests (Fig. 2-e and 2-f), both the treatment group and treatment timing are randomly assigned.

defined as the natural logarithm of one plus the total number of crash weeks in a given fiscal year. The results, reported in Appendix A.4, show that the coefficients on  $Treated \times Post$  remain significantly positive, reinforcing the robustness of our findings.

**3.2.2.5. Alternative measures of AI investment.** As an additional robustness check, we replace the AI pilot zone policy indicator with firm-level measures of actual AI investment to verify the consistency of our findings. Specifically, we construct two alternative measures,  $AI\_level$  and  $\ln AI$ , to re-estimate the baseline regression.  $AI\_level$  is defined as the ratio of AI investment to total assets, while  $\ln AI$  is the natural logarithm of firms' AI-related expenditure.

The results, presented in Appendix A.5, show that in all specifications the coefficients on  $AI\_level$  and  $\ln AI$  remain significantly positive. This indicates that firms with higher levels of AI investment are more likely to experience stock price crash risk. These findings are consistent with our main results and further reinforce the robustness of the relationship between AI investment and crash risk.

**3.2.2.6. Control for firm-fixed effects.** Considering that our baseline specification controls only for industry-fixed effects, we further



**Table 3**

Tests of the information transparency channel.

|                     | (1)                  | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   |
|---------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                     | $DisAcc_t$           | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         | $AuditOp_t$           | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         |
| $Treat \times Post$ | 0.006***<br>(2.58)   |                       |                       | 0.010**<br>(2.15)     |                       |                       |
| $DisAcc_t$          |                      | 0.128**<br>(2.14)     | 0.100***<br>(2.59)    |                       |                       |                       |
| $AuditOp_t$         |                      |                       |                       |                       | 0.168***<br>(5.41)    | 0.122***<br>(5.71)    |
| $NCSKEW_t$          |                      | 0.042***<br>(5.95)    |                       |                       | 0.056***<br>(7.98)    |                       |
| $DUVOL_t$           |                      |                       | 0.042***<br>(6.13)    |                       |                       | 0.041***<br>(6.09)    |
| $Size_t$            | 0.001<br>(0.46)      | 0.072***<br>(12.60)   | 0.036***<br>(9.69)    | −0.001<br>(−0.49)     | 0.068***<br>(12.36)   | 0.036***<br>(9.66)    |
| $Lev_t$             | −0.002<br>(−0.36)    | −0.058*<br>(−1.67)    | −0.042*<br>(−1.92)    | 0.080***<br>(5.09)    | −0.066**<br>(−1.99)   | −0.047**<br>(−2.14)   |
| $ROA_t$             | 0.592***<br>(45.84)  | −0.268***<br>(−2.53)  | −0.206***<br>(−3.03)  | −0.738***<br>(−15.91) | −0.072<br>(−0.73)     | −0.086<br>(−1.34)     |
| $PAF_t$             | −0.035***<br>(−6.92) | 0.008<br>(0.19)       | −0.020<br>(−0.77)     | −0.035***<br>(−2.58)  | 0.011<br>(0.27)       | −0.021<br>(−0.80)     |
| $SOE_t$             | 0.003*<br>(1.79)     | −0.059***<br>(−4.52)  | −0.044***<br>(−5.33)  | −0.024***<br>(−6.20)  | −0.050***<br>(−4.15)  | −0.041***<br>(−4.97)  |
| $Growth_t$          | −0.003<br>(−1.06)    | −0.006<br>(−0.35)     | −0.001<br>(−0.10)     | −0.014***<br>(−2.87)  | −0.007<br>(−0.41)     | 0.002<br>(0.21)       |
| $Top1_t$            | −0.014***<br>(−2.93) | −0.031<br>(−0.79)     | 0.006<br>(0.22)       | −0.020**<br>(−2.04)   | −0.027<br>(−0.73)     | 0.007<br>(0.29)       |
| $Mshare_t$          | 0.010<br>(1.22)      | 0.175***<br>(2.72)    | 0.096**<br>(2.31)     | −0.028*<br>(−1.78)    | 0.164***<br>(2.68)    | 0.098**<br>(2.36)     |
| $Duality_t$         | 0.001<br>(0.60)      | 0.017<br>(1.21)       | 0.008<br>(0.95)       | −0.001<br>(−0.36)     | 0.016<br>(1.23)       | 0.009<br>(1.07)       |
| $Separation_t$      | −0.008<br>(−0.98)    | −0.054<br>(−0.76)     | −0.016<br>(−0.35)     | 0.0031<br>(0.14)      | −0.026<br>(−0.38)     | −0.017<br>(−0.37)     |
| $Intensity_t$       | 0.002***<br>(6.59)   | −0.001<br>(−0.22)     | −0.001<br>(−0.02)     | 0.006***<br>(5.23)    | −0.001<br>(−0.24)     | 0.001<br>(0.07)       |
| $BM_t$              | 0.016***<br>(4.25)   | −0.433***<br>(−14.68) | −0.244***<br>(−12.80) | −0.078***<br>(−7.68)  | −0.397***<br>(−14.11) | −0.239***<br>(−12.61) |
| $RET_t$             | −0.231**<br>(−2.29)  | 12.050***<br>(14.45)  | 8.537***<br>(15.27)   | −1.858***<br>(−7.97)  | 12.650***<br>(15.39)  | 8.685***<br>(15.66)   |
| $Sigma_t$           | 0.097***<br>(2.59)   | −0.998***<br>(−2.99)  | −1.163***<br>(−5.43)  | 0.326***<br>(3.44)    | −0.994***<br>(−3.03)  | −1.155***<br>(−5.44)  |
| $Turnover_t$        | −0.001**<br>(−2.03)  | −0.001***<br>(−5.92)  | −0.001***<br>(−5.94)  | 0.001***<br>(5.61)    | −0.001***<br>(−6.11)  | −0.001***<br>(−6.14)  |
| _cons               | −0.032**<br>(−2.14)  | −1.662***<br>(−13.76) | −0.844***<br>(−10.83) | 0.092**<br>(2.28)     | −1.624***<br>(−13.53) | −0.849***<br>(−10.95) |
| Industry FE         | Y                    | Y                     | Y                     | Y                     | Y                     | Y                     |
| Year FE             | Y                    | Y                     | Y                     | Y                     | Y                     | Y                     |
| City FE             | Y                    | Y                     | Y                     | Y                     | Y                     | Y                     |
| N                   | 25,472               | 25,472                | 25,472                | 25,744                | 25,744                | 25,732                |
| adj. R <sup>2</sup> | 0.26                 | 0.06                  | 0.06                  | 0.14                  | 0.06                  | 0.06                  |

This table reports the results as to the test of the information channel through which AI investment increases stock price crash risk. Column (1) reports the results of the regression of earnings management ( $DisAcc$ ) on AI investment reform ( $Treated \times Post$ ). Columns (2) and (3) report the results of the baseline regression that is augmented by  $DisAcc$  but excludes  $Treated \times Post$ . Column (4) reports the results of the regression of auditor opinion ( $AuditOp$ ) on AI investment reform ( $Treated \times Post$ ). Columns (5) and (6) report the results of the baseline regression that is augmented by  $AuditOp$  but excludes  $Treated \times Post$ . Robust t-values clustered at the firm level are reported in parentheses. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

replace them with firm-fixed effects to mitigate concerns about unobserved firm-specific characteristics that may influence stock price crash risk. The results, reported in [Appendix A.6](#), show that both coefficients on  $Treat \times Post$  remain significantly positive. These findings further confirm the robustness of our baseline DID estimates, alleviating concerns that the observed effects are driven by time-invariant firm-level heterogeneity.

**3.2.2.7. Clustering standard errors at the city level.** To further test the robustness of our baseline results, we re-estimate the regressions by clustering standard errors at the city level instead of the firm level. This adjustment accounts for potential correlations among firms located in the same city, which may be exposed to similar regional shocks, local economic conditions, or policy environments. [Appendix A.7](#) presents the results. The coefficients on  $Treat \times Post$  remain significantly positive and are qualitatively consistent with our baseline estimates, indicating that our findings are not sensitive to the choice of clustering level.

**Table 4**

Tests of expectation management channel.

|                     | (1)                  | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   |
|---------------------|----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                     | $EX\_Tone1_t$        | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         | $EX\_Tone2_t$         | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         |
| $Treat \times Post$ | 0.001**<br>(2.03)    |                       |                       | 0.002**<br>(1.98)     |                       |                       |
| $EX\_Tone1_t$       |                      | 0.249**<br>(2.40)     | 0.241***<br>(2.59)    |                       |                       |                       |
| $EX\_Tone2_t$       |                      |                       |                       |                       | 0.019**<br>(2.22)     | 0.006**<br>(2.12)     |
| $NCSKEW_t$          |                      | 0.041***<br>(5.81)    |                       |                       | 0.041***<br>(5.81)    |                       |
| $DUVOL_t$           |                      |                       | 0.042***<br>(6.11)    |                       |                       | 0.042***<br>(6.11)    |
| $Size_t$            | 0.001***<br>(6.21)   | 0.073***<br>(12.58)   | 0.036***<br>(9.49)    | 0.008***<br>(17.84)   | 0.072***<br>(12.63)   | 0.036***<br>(9.55)    |
| $Lev_t$             | -0.004***<br>(-6.36) | -0.061*<br>(-1.76)    | -0.039*<br>(-1.79)    | -0.039***<br>(-15.46) | -0.062*<br>(-1.80)    | -0.041*<br>(-1.85)    |
| $ROA_t$             | 0.014***<br>(9.40)   | -0.228**<br>(-2.24)   | -0.164**<br>(-2.51)   | 0.099***<br>(13.60)   | -0.222**<br>(-2.19)   | -0.160**<br>(-2.45)   |
| $PAF_t$             | 0.001<br>(0.23)      | 0.004<br>(0.10)       | -0.025<br>(-0.96)     | 0.006*<br>(1.82)      | 0.004<br>(0.10)       | -0.025<br>(-0.95)     |
| $SOE_t$             | -0.003***<br>(-8.03) | -0.058***<br>(-4.45)  | -0.043***<br>(-5.14)  | -0.019***<br>(-20.69) | -0.059***<br>(-4.52)  | -0.043***<br>(-5.22)  |
| $Growth_t$          | 0.001***<br>(5.59)   | -0.001<br>(-0.04)     | 0.003<br>(0.33)       | 0.009***<br>(7.25)    | -0.001<br>(-0.01)     | 0.004<br>(0.36)       |
| $Top1_t$            | -0.001<br>(-0.56)    | -0.020<br>(-0.49)     | 0.012<br>(0.48)       | -0.004<br>(-1.47)     | -0.020<br>(-0.50)     | 0.012<br>(0.48)       |
| $Mshare_t$          | 0.005***<br>(4.19)   | 0.179***<br>(2.81)    | 0.096**<br>(2.31)     | 0.044***<br>(9.07)    | 0.181***<br>(2.84)    | 0.097**<br>(2.35)     |
| $Duality_t$         | 0.001<br>(0.61)      | 0.016<br>(1.14)       | 0.009<br>(1.06)       | 0.001<br>(0.91)       | 0.016<br>(1.14)       | 0.009<br>(1.07)       |
| $Separation_t$      | -0.004**<br>(-2.40)  | -0.067<br>(-0.94)     | -0.025<br>(-0.55)     | -0.049***<br>(-9.59)  | -0.068<br>(-0.97)     | -0.027<br>(-0.58)     |
| $Intensity_t$       | -0.001<br>(-1.56)    | 0.001<br>(0.23)       | 0.001<br>(0.65)       | -0.001***<br>(-4.32)  | 0.001<br>(0.22)       | 0.001<br>(0.63)       |
| $BM_t$              | -0.001*<br>(-1.83)   | -0.438***<br>(-14.90) | -0.245***<br>(-12.86) | -0.014***<br>(-6.24)  | -0.438***<br>(-14.92) | -0.245***<br>(-12.87) |
| $RET_t$             | 0.014*<br>(1.72)     | 11.820***<br>(14.17)  | 8.375***<br>(14.94)   | 0.089<br>(1.27)       | 11.82***<br>(14.17)   | 8.379***<br>(14.94)   |
| $Sigma_t$           | -0.014***<br>(-3.17) | -0.871***<br>(-2.61)  | -1.082***<br>(-5.04)  | -0.102***<br>(-3.97)  | -0.877***<br>(-2.62)  | -1.086***<br>(-5.05)  |
| $Turnover_t$        | -0.001***<br>(-2.98) | -0.001***<br>(-5.99)  | -0.001***<br>(-5.96)  | -0.001***<br>(-3.27)  | -0.001***<br>(-6.00)  | -0.001***<br>(-5.97)  |
| _cons               | -0.016***<br>(-5.47) | -1.676***<br>(-13.82) | -0.841***<br>(-10.75) | -0.147***<br>(-16.46) | -1.683***<br>(-13.86) | -0.846***<br>(-10.80) |
| Industry FE         | Y                    | Y                     | Y                     | Y                     | Y                     | Y                     |
| Year FE             | Y                    | Y                     | Y                     | Y                     | Y                     | Y                     |
| City FE             | Y                    | Y                     | Y                     | Y                     | Y                     | Y                     |
| N                   | 25,354               | 25,354                | 25,354                | 25,366                | 25,354                | 25,354                |
| adj. R <sup>2</sup> | 0.16                 | 0.06                  | 0.06                  | 0.20                  | 0.06                  | 0.06                  |

This table reports the results as to the test of the expectation management channel through which AI investment increases stock price crash risk. Column (1) and Column (4) report the results of the regression of over-optimism ( $EX\_Tone1$  and  $EX\_Tone2$ ) on AI investment reform ( $Treated \times Post$ ). Columns (2) and (3) report the results of the baseline regression that is augmented by  $EX\_Tone1$  but excludes  $Treated \times Post$ . Columns (5) and (6) report the results of the baseline regression that is augmented by  $EX\_Tone2$  but excludes  $Treated \times Post$ . Robust t-values clustered at the firm level are reported in parentheses. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

### 3.2.3. Mechanism analysis

**3.2.3.1. Tests of information transparency channel.** As discussed in the registered study (Bai and Zhao, 2025), AI investment may increase stock price crash risk by deteriorating the information environment. The inherent complexity, algorithmic opacity, and discretionary nature of AI systems may provide managers with greater opportunities to manipulate financial information or conceal negative news.

To empirically examine the information transparency channel, we use earnings management ( $DisAcc$ ) and audit opinion ( $AuditOp$ ) as proxies for information transparency.  $DisAcc$  represents a firm's abnormal accruals in a fiscal year, estimated using the modified Jones model. Higher levels of discretionary accruals reflect more aggressive earnings manipulation, thereby indicating a less transparent information environment.  $AuditOp$  reflects the opinion issued by the external auditor on the firm's financial statements, taking a value of 1 for a non-standard audit opinion, and 0 for a standard opinion. In summary, higher values of  $DisAcc$  or  $AuditOp$  reflect lower



**Table 5**  
Effect on firm value.

|                                              | (1)                         | (2)                         | (3)                         | (4)                     | (5)                     | (6)                     |
|----------------------------------------------|-----------------------------|-----------------------------|-----------------------------|-------------------------|-------------------------|-------------------------|
|                                              | <i>TobinQ<sub>t+1</sub></i> | <i>TobinQ<sub>t+1</sub></i> | <i>TobinQ<sub>t+1</sub></i> | <i>PB<sub>t+1</sub></i> | <i>PB<sub>t+1</sub></i> | <i>PB<sub>t+1</sub></i> |
| <i>Treated</i> × <i>Post</i>                 | 0.100***<br>(2.70)          | 0.135***<br>(3.19)          | 0.127***<br>(3.00)          | 0.102***<br>(2.68)      | 0.139***<br>(3.15)      | 0.131***<br>(2.97)      |
| <i>NCSKEW</i>                                |                             | 0.140***<br>(12.61)         |                             |                         | 0.144***<br>(12.40)     |                         |
| <i>DUVOL</i>                                 |                             |                             | 0.206***<br>(11.55)         |                         |                         | 0.212***<br>(11.35)     |
| <i>Treated</i> × <i>Post</i> × <i>NCSKEW</i> |                             | 0.071***<br>(2.69)          |                             |                         | 0.073***<br>(2.66)      |                         |
| <i>Treated</i> × <i>Post</i> × <i>DUVOL</i>  |                             |                             | 0.085**<br>(2.03)           |                         |                         | 0.089**<br>(2.05)       |
| <i>Size<sub>t</sub></i>                      | −0.380***<br>(−18.05)       | −0.379***<br>(−18.20)       | −0.379***<br>(−18.16)       | −0.389***<br>(−17.16)   | −0.388***<br>(−17.29)   | −0.388***<br>(−17.26)   |
| <i>Lev<sub>t</sub></i>                       | −0.060<br>(−0.55)           | −0.067<br>(−0.62)           | −0.064<br>(−0.59)           | −0.034<br>(−0.30)       | −0.042<br>(−0.37)       | −0.039<br>(−0.34)       |
| <i>ROA<sub>t</sub></i>                       | 2.387***<br>(7.39)          | 2.301***<br>(7.25)          | 2.305***<br>(7.26)          | 2.460***<br>(7.28)      | 2.370***<br>(7.14)      | 2.375***<br>(7.14)      |
| <i>PAF<sub>t</sub></i>                       | −0.329***<br>(−3.35)        | −0.324***<br>(−3.34)        | −0.322***<br>(−3.31)        | −0.341***<br>(−3.33)    | −0.336***<br>(−3.32)    | −0.334***<br>(−3.30)    |
| <i>SOE<sub>t</sub></i>                       | −0.051<br>(−1.47)           | −0.036<br>(−1.05)           | −0.037<br>(−1.09)           | −0.051<br>(−1.42)       | −0.036<br>(−1.00)       | −0.037<br>(−1.04)       |
| <i>Growth<sub>t</sub></i>                    | −0.023<br>(−0.80)           | −0.038<br>(−1.35)           | −0.034<br>(−1.21)           | −0.023<br>(−0.76)       | −0.039<br>(−1.30)       | −0.035<br>(−1.16)       |
| <i>Top1<sub>t</sub></i>                      | 0.304***<br>(3.25)          | 0.304***<br>(3.27)          | 0.300***<br>(3.23)          | 0.299***<br>(3.11)      | 0.299***<br>(3.13)      | 0.295***<br>(3.08)      |
| <i>Mshare<sub>t</sub></i>                    | −0.851***<br>(−4.93)        | −0.872***<br>(−5.11)        | −0.872***<br>(−5.10)        | −0.878***<br>(−4.94)    | −0.899***<br>(−5.12)    | −0.899***<br>(−5.11)    |
| <i>Duality<sub>t</sub></i>                   | 0.133***<br>(3.72)          | 0.132***<br>(3.71)          | 0.132***<br>(3.72)          | 0.135***<br>(3.58)      | 0.133***<br>(3.57)      | 0.134***<br>(3.57)      |
| <i>Separation<sub>t</sub></i>                | −0.021<br>(−0.12)           | −0.024<br>(−0.14)           | −0.033<br>(−0.19)           | −0.018<br>(−0.10)       | −0.021<br>(−0.12)       | −0.030<br>(−0.17)       |
| <i>Intensity<sub>t</sub></i>                 | 0.012**<br>(2.13)           | 0.012**<br>(2.18)           | 0.012**<br>(2.15)           | 0.013**<br>(2.21)       | 0.013**<br>(2.26)       | 0.013**<br>(2.23)       |
| <i>RET<sub>t</sub></i>                       | 29.960***<br>(18.75)        | 31.810***<br>(19.54)        | 32.360***<br>(19.55)        | 30.810***<br>(18.35)    | 32.720***<br>(19.12)    | 33.280***<br>(19.12)    |
| <i>Sigma<sub>t</sub></i>                     | −0.274<br>(−0.48)           | 0.947*<br>(1.67)            | 0.649<br>(1.14)             | −0.421<br>(−0.71)       | 0.838<br>(1.43)         | 0.530<br>(0.90)         |
| <i>Turnover<sub>t</sub></i>                  | 0.046**<br>(2.04)           | 0.039*<br>(1.75)            | 0.040*<br>(1.81)            | 0.047**<br>(2.02)       | 0.040*<br>(1.74)        | 0.041*<br>(1.80)        |
| _cons                                        | 10.310***<br>(22.92)        | 10.270***<br>(23.09)        | 10.300***<br>(23.07)        | 10.510***<br>(21.71)    | 10.470***<br>(21.87)    | 10.500***<br>(21.85)    |
| <i>Industry FE</i>                           | Y                           | Y                           | Y                           | Y                       | Y                       | Y                       |
| <i>Year FE</i>                               | Y                           | Y                           | Y                           | Y                       | Y                       | Y                       |
| <i>City FE</i>                               | Y                           | Y                           | Y                           | Y                       | Y                       | Y                       |
| <i>N</i>                                     | 21,814                      | 21,814                      | 21,814                      | 21,814                  | 21,814                  | 21,814                  |
| <i>adj. R<sup>2</sup></i>                    | 0.36                        | 0.37                        | 0.37                        | 0.35                    | 0.36                    | 0.36                    |

This table reports the results of how AI pilot zone policy affects the firm value. Column (1) and Column (4) report the results of the regression of the policy (*Treated*×*Post*) on firm value (*TobinQ* and *PB*). Columns (2) and (3) report the results of the baseline regression that is augmented by *EX\_Tone1* but excludes *Treated*×*Post*. Columns (5) and (6) report the results of the baseline regression that is extended using Eq. (8) by including the interaction terms between *Treat*×*Post* and stock price crash risk. Robust t-values clustered at the firm level are reported in parentheses. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

levels of information transparency. We perform the mediation analysis by running the following regressions:

$$DisAcc_{i,t} \text{ or } AuditOp_{i,t} = \beta_0 + \beta_1 Treat_t \times Post_t + \beta_2 Controls_{i,t} + Industry + Year + City + \varepsilon_{i,t} \quad (3)$$

$$NCSKEW_{i,t+1} \text{ or } DUVOL_{i,t+1} = \beta_0 + \beta_1 DisAcc_{i,t} \text{ or } AuditOp_{i,t} + \beta_2 Controls_{i,t} + Industry + Year + City + \varepsilon_{i,t} \quad (4)$$

where the mediator variables are *DisAcc* and *AuditOp*. If the mediating effect exists, the coefficients of *Treated*×*Post* for *DisAcc* and *AuditOp* in Eq.(3) should be positive and statistically significant at conventional levels, and their coefficients in Eq. (4) should also be significantly positive.

Table 3 presents the results of the mechanism tests for the information transparency channel. Both the coefficients of *Treated*×*Post* in the first-stage regressions (reported in Columns(1) and (4)) and those of the mediator (*DisAcc* and *AuditOp*) in the second-stage regression (reported in Columns (2),(3),(5),(6)) are significantly positive. These findings support the hypothesis that AI investment increases stock price crash risk by reducing information transparency.

**Table 6**

The moderating effect of information transparency.

|                                                   | (1)                   | (2)                   | (3)                   | (4)                   |
|---------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                                   | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         |
| <i>Treated</i> × <i>Post</i>                      | 0.005<br>(0.20)       | 0.007<br>(0.45)       | 0.019<br>(0.92)       | 0.019<br>(1.46)       |
| <i>DomesticCPA</i>                                | −0.017<br>(−1.39)     | −0.002<br>(−0.28)     |                       |                       |
| <i>Treated</i> × <i>Post</i> × <i>DomesticCPA</i> | 0.058**<br>(2.26)     | 0.036**<br>(2.20)     |                       |                       |
| <i>Int_Assets</i>                                 |                       |                       | −0.075<br>(−0.82)     | −0.063<br>(−1.02)     |
| <i>Treated</i> × <i>Post</i> × <i>Int_Assets</i>  |                       |                       | 0.518***<br>(2.61)    | 0.245*<br>(1.91)      |
| <i>NCSKEW<sub>t</sub></i>                         | 0.042***<br>(5.95)    |                       | 0.042***<br>(5.95)    |                       |
| <i>DUVOL<sub>t</sub></i>                          |                       | 0.042***<br>(6.14)    |                       | 0.042***<br>(6.16)    |
| <i>Size<sub>t</sub></i>                           | 0.072***<br>(12.51)   | 0.035***<br>(9.40)    | 0.072***<br>(12.54)   | 0.035***<br>(9.56)    |
| <i>Lev<sub>t</sub></i>                            | −0.056<br>(−1.63)     | −0.038*<br>(−1.71)    | −0.056<br>(−1.61)     | −0.038*<br>(−1.72)    |
| <i>ROA<sub>t</sub></i>                            | −0.239**<br>(−2.41)   | −0.178***<br>(−2.78)  | −0.236**<br>(−2.38)   | −0.175***<br>(−2.74)  |
| <i>PAF<sub>t</sub></i>                            | 0.003<br>(0.07)       | −0.023<br>(−0.89)     | 0.006<br>(0.13)       | −0.023<br>(−0.88)     |
| <i>SOE<sub>t</sub></i>                            | −0.057***<br>(−4.44)  | −0.043***<br>(−5.23)  | −0.057***<br>(−4.45)  | −0.043***<br>(−5.24)  |
| <i>Growth<sub>t</sub></i>                         | −0.005<br>(−0.31)     | 0.001<br>(0.02)       | −0.005<br>(−0.29)     | 0.001<br>(0.03)       |
| <i>Top1<sub>t</sub></i>                           | −0.031<br>(−0.80)     | 0.003<br>(0.12)       | −0.032<br>(−0.82)     | 0.004<br>(0.18)       |
| <i>Mshare<sub>t</sub></i>                         | 0.174***<br>(2.70)    | 0.091**<br>(2.19)     | 0.171***<br>(2.66)    | 0.092**<br>(2.20)     |
| <i>Duality<sub>t</sub></i>                        | 0.017<br>(1.23)       | 0.009<br>(1.04)       | 0.016<br>(1.21)       | 0.009<br>(1.03)       |
| <i>Separation<sub>t</sub></i>                     | −0.051<br>(−0.74)     | −0.018<br>(−0.38)     | −0.053<br>(−0.76)     | −0.017<br>(−0.37)     |
| <i>Intensity<sub>t</sub></i>                      | 0.001<br>(0.23)       | 0.001<br>(0.66)       | 0.001<br>(0.20)       | 0.001<br>(0.65)       |
| <i>BM<sub>t</sub></i>                             | −0.440***<br>(−15.05) | −0.247***<br>(−13.03) | −0.442***<br>(−15.12) | −0.248***<br>(−13.10) |
| <i>RET<sub>t</sub></i>                            | 12.02***<br>(14.52)   | 8.489***<br>(15.28)   | 11.94***<br>(14.41)   | 8.455***<br>(15.21)   |
| <i>Sigma<sub>t</sub></i>                          | −0.921***<br>(−2.78)  | −1.101***<br>(−5.18)  | −0.905***<br>(−2.73)  | −1.096***<br>(−5.16)  |
| <i>Turnover<sub>t</sub></i>                       | −0.001***<br>(−5.95)  | −0.001***<br>(−5.95)  | −0.001***<br>(−5.95)  | −0.001***<br>(−5.97)  |
| <i>_cons</i>                                      | −1.651***<br>(−13.67) | −0.828***<br>(−10.65) | −1.654***<br>(−13.71) | −0.836***<br>(−10.75) |
| <i>Industry FE</i>                                | Y                     | Y                     | Y                     | Y                     |
| <i>Year FE</i>                                    | Y                     | Y                     | Y                     | Y                     |
| <i>City FE</i>                                    | Y                     | Y                     | Y                     | Y                     |
| <i>N</i>                                          | 25,732                | 25,732                | 25,732                | 25,732                |
| <i>adj. R<sup>2</sup></i>                         | 0.06                  | 0.06                  | 0.06                  | 0.06                  |

This table reports the results for the moderating effect of information transparency on the association between AI pilot zone policy and stock price crash risk (*NCSKEW* and *DUVOL*). The moderating effect is captured by the interaction term between the indicator for information transparency (*DomesticCPA* and *Int\_Assets*) and *Treat*×*Post*. *DomesticCPA* equals 1 if the firm engages a Chinese domestic CPA firm as its auditor, and 0 otherwise. Column (1) and (2) report the moderating effect on *DomesticCPA*. *Int\_Assets* is the ratio of intangible assets to total assets. Column (3) and (4) report the moderating effect on *Int\_Assets*. Robust t-values clustered at the firm level are reported in parentheses. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

**3.2.3.2. Tests of expectation management channel.** In this section, we examine whether expectation management serves as another underlying mechanism through which AI investment affects stock price crash risk. Since AI investment is often presented as part of a firm's long-term strategic vision, it provides managers with a convenient narrative to shape investor perceptions. This may lead to the strategic use of optimistic language in corporate disclosures, overstating future growth potential while simultaneously concealing negative information.

To test this channel, we use *EX\_Tone1* and *EX\_Tone2* as proxies for managerial expectation management. These variables capture the extent to which the optimistic tone in a firm's annual financial report exceeds the industry average in a given year, thereby reflecting

**Table 7**

The moderating effect of resource constraint.

|                                            | (1)                   | (2)                   | (3)                   | (4)                   |
|--------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                            | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         |
| <i>Treated</i> × <i>Post</i>               | 0.099***<br>(3.91)    | 0.063***<br>(3.84)    | −0.024<br>(−1.30)     | −0.008<br>(−0.68)     |
| <i>Cash</i>                                | 0.097*<br>(1.83)      | 0.064*<br>(1.86)      |                       |                       |
| <i>Treated</i> × <i>Post</i> × <i>Cash</i> | −0.311***<br>(−3.19)  | −0.179***<br>(−2.79)  |                       |                       |
| <i>HHI</i>                                 |                       |                       | 0.040<br>(0.38)       | 0.003<br>(0.04)       |
| <i>Treated</i> × <i>Post</i> × <i>HHI</i>  |                       |                       | 0.267**<br>(2.13)     | 0.156*<br>(1.95)      |
| <i>NCSKEW<sub>t</sub></i>                  | 0.042***<br>(5.95)    |                       | 0.059***<br>(9.12)    |                       |
| <i>DUVOL<sub>t</sub></i>                   |                       | 0.042***<br>(6.14)    |                       | 0.058***<br>(9.03)    |
| <i>Size<sub>t</sub></i>                    | 0.072***<br>(12.57)   | 0.036***<br>(9.57)    | 0.065***<br>(12.05)   | 0.031***<br>(9.09)    |
| <i>Lev<sub>t</sub></i>                     | −0.050<br>(−1.44)     | −0.034<br>(−1.50)     | −0.046<br>(−1.48)     | −0.031<br>(−1.57)     |
| <i>ROA<sub>t</sub></i>                     | −0.242**<br>(−2.43)   | −0.179***<br>(−2.79)  | −0.168*<br>(−1.87)    | −0.138**<br>(−2.40)   |
| <i>PAF<sub>t</sub></i>                     | 0.014<br>(0.34)       | −0.015<br>(−0.55)     | 0.035<br>(0.93)       | −0.003<br>(−0.14)     |
| <i>SOE<sub>t</sub></i>                     | −0.057***<br>(−4.46)  | −0.044***<br>(−5.27)  | −0.059***<br>(−5.24)  | −0.041***<br>(−5.74)  |
| <i>Growth<sub>t</sub></i>                  | −0.004<br>(−0.27)     | 0.001<br>(0.05)       | −0.008<br>(−0.54)     | −0.002<br>(−0.24)     |
| <i>Top1<sub>t</sub></i>                    | −0.031<br>(−0.78)     | 0.005<br>(0.19)       | −0.036<br>(−1.05)     | −0.001<br>(−0.00)     |
| <i>Mshare<sub>t</sub></i>                  | 0.177***<br>(2.75)    | 0.094**<br>(2.26)     | 0.149**<br>(2.48)     | 0.077**<br>(2.00)     |
| <i>Duality<sub>t</sub></i>                 | 0.015<br>(1.11)       | 0.008<br>(0.95)       | 0.015<br>(1.22)       | 0.009<br>(1.15)       |
| <i>Separation<sub>t</sub></i>              | −0.053<br>(−0.77)     | −0.018<br>(−0.39)     | 0.001<br>(0.00)       | 0.012<br>(0.29)       |
| <i>Intensity<sub>t</sub></i>               | 0.001<br>(0.23)       | 0.001<br>(0.68)       | 0.001<br>(0.15)       | 0.001<br>(0.57)       |
| <i>BM<sub>t</sub></i>                      | −0.442***<br>(−15.13) | −0.248***<br>(−13.11) | −0.392***<br>(−14.06) | −0.219***<br>(−12.34) |
| <i>RET<sub>t</sub></i>                     | 11.940***<br>(14.40)  | 8.443***<br>(15.17)   | 12.550***<br>(14.38)  | 8.858***<br>(15.83)   |
| <i>Sigma<sub>t</sub></i>                   | −0.914***<br>(−2.76)  | −1.099***<br>(−5.17)  | −0.940***<br>(−2.89)  | −1.162***<br>(−5.62)  |
| <i>Turnover<sub>t</sub></i>                | −0.001***<br>(−5.92)  | −0.001***<br>(−5.93)  | −0.001***<br>(−6.04)  | −0.001***<br>(−5.75)  |
| <i>_cons</i>                               | −1.683***<br>(−13.82) | −0.854***<br>(−10.91) | −1.538***<br>(−13.78) | −0.760***<br>(−10.66) |
| <i>Industry FE</i>                         | Y                     | Y                     | Y                     | Y                     |
| <i>Year FE</i>                             | Y                     | Y                     | Y                     | Y                     |
| <i>City FE</i>                             | Y                     | Y                     | Y                     | Y                     |
| <i>N</i>                                   | 25,732                | 25,732                | 25,744                | 25,744                |
| <i>adj. R<sup>2</sup></i>                  | 0.06                  | 0.06                  | 0.06                  | 0.06                  |

This table reports the results for the moderating effect of resource constraint on the association between AI pilot zone policy and stock price crash risk (*NCSKEW* and *DUVOL*). The moderating effect is captured by the interaction term between the indicator for information transparency (*Cash* and *HHI*) and *Treat*×*Post*. *Cash* is the ratio of cash and cash equivalents to total assets, reflecting a firm's internal liquidity. Column (1) and (2) report the moderating effect on *Cash*. *HHI* refers to the Herfindahl-Hirschman Index, calculated as the sum of the squared market shares of firms within the same industry. Column (3) and (4) report the moderating effect on *HHI*. Robust t-values clustered at the firm level are reported in parentheses. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

relative over-optimism. The underlying tone measures are derived from the textual analysis of annual report disclosures using the Loughran and McDonald (LM) dictionary. Specifically, *LM\_tone1* is calculated as 100 times the difference between the number of positive and negative words divided by the total number of words in the report, while *LM\_tone2* is defined as 100 times the difference between the number of positive and negative words divided by the sum of positive and negative words. The raw tone data are obtained from the Annual Report Tone Database provided by the China Stock Market & Accounting Research (CNRDS) platform. We construct *EX\_tone1* and *EX\_tone2* by demeaning *LM\_tone1* and *LM\_tone2* at the industry-year level. These excess tone variables are then used in the mediation analysis to assess whether expectation management serves as a channel through which AI investment affects stock price

crash risk. We perform the mediation analysis by running the following regressions:

$$EX\_tone1_{i,t} \text{ or } EX\_tone2_{i,t} = \beta_0 + \beta_1 Treat_i \times Post_i + \beta_2 Controls_{i,t} + Industry + Year + City + \varepsilon_{i,t} \quad (5)$$

$$NCSKEW_{i,t+1} \text{ or } DUVOL_{i,t+1} = \beta_0 + \beta_1 EX\_tone1_{i,t} \text{ or } EX\_tone2_{i,t} + \beta_2 Controls_{i,t} + Industry + Year + City + \varepsilon_{i,t} \quad (6)$$

where the mediator variables are *EX\_tone1* and *EX\_tone2*. If the mediating effect exists, the coefficients of *Treated* × *Post* for *EX\_tone1* and *EX\_tone2* in Eq. (5) should be positive and statistically significant at conventional levels, and their coefficients in Eq. (6) should also be significantly positive.

Table 4 reports the results of the mechanism tests for the expectation management channel. The coefficients of *Treated* × *Post* in the first-stage regressions (reported in Columns (1) and (4)) are significantly positive, indicating that the AI pilot zone policy leads firms to adopt a more optimistic disclosure tone. Moreover, the coefficients of the mediators (*EX\_tone1* and *EX\_tone2*) in the second-stage regressions (reported in Columns (2), (3), (5), and (6)) are also significantly positive. These results indicate that heightened managerial optimism acts as an important transmission channel through which AI investment contributes to increased stock price crash risk.

### 3.2.4. Additional tests

**3.2.4.1. Effect on firm value.** In the preceding sections, we examined how AI investment influences stock price crash risk. A natural follow-up question is whether the stock price crash risk induced by AI investment further affects firm value. To explore this issue, we extend our analysis to investigate the economic consequence of AI investment on firm value. Empirically, we perform regressions following the specifications below:

$$Firm\ Value_{i,t+1} = \beta_0 + \beta_1 Treat_i \times Post_i + \beta_2 Controls_{i,t} + Industry + Year + City + \varepsilon_{i,t} \quad (7)$$

$$Firm\ Value_{i,t+1} = \beta_0 + \beta_1 Treat_i \times Post_i + \beta_2 NCSKEW_{i,t} \text{ or } DUVOL_{i,t} + \beta_3 Treat_i \times Post_i \times NCSKEW_{i,t} \text{ or } DUVOL_{i,t} + \beta_4 Controls_{i,t} + Industry + Year + City + \varepsilon_{i,t} \quad (8)$$

where Firm Value is measured by Tobin's Q and the market-to-book ratio. All other variables are defined in the previous sections.

In Column (1) and (4) of Table 5, we estimate Eq. (7) and find that the coefficients of *Treat* × *Post* are significantly positive at the 1 % level, indicating that firm value increases following the pilot AI-zone policy. To further explore the role of stock price crash risk, we extend our analysis using Eq. (8) by including the interaction terms between *Treat* × *Post* and stock price crash risk. The results in Column (2), (3), (5), and (6) show that the coefficients of *Treat* × *Post* × *NCSKEW* and *Treat* × *Post* × *DUVOL* are also significantly positive. These findings suggest an environment of excessive optimism, where the market initially overreacts to the perceived benefits of AI adoption, thereby inflating firm valuations. Meanwhile, the heightened crash risk reveals the hidden downside of such mispricing and managerial incentive to withhold bad news during the policy implementation.

**3.2.4.2. Cross-sectional analyses.** In this section, we explore how the baseline results vary across different firm characteristics. Specifically, we investigate whether information transparency moderates the relationship between the AI pilot policy and stock price crash risk. We use two proxies for firm-level information transparency: whether a firm is audited by a domestic accounting firm (*DomesticCPA*), and the ratio of intangible assets to total assets (*Int\_Assets*). These variables are interacted with the AI policy treatment indicator to assess their moderating effects.

Table 6 reports the results. The coefficients on the interaction terms between the AI policy indicator and both proxies for lower information transparency are significantly positive. This indicates that the positive effect of the AI policy on stock price crash risk is more pronounced among firms with lower transparency. Overall, these findings suggest that weaker external monitoring and more opaque asset structures amplify the adverse effects of AI policy shocks, underscoring the critical role of information asymmetry in shaping market reactions.

We further investigate how the baseline results vary across firm and industry characteristics related to resource constraints. Specifically, we examine whether the impact of the AI pilot policy on stock price crash risk is moderated by firms' internal financial resources and external competitive environment. To test this, we introduce interaction terms involving *Cash* and *HHI* into the baseline regression. *Cash* is the ratio of cash and cash equivalents to total assets, capturing a firm's internal liquidity position. *HHI* denotes the Herfindahl-Hirschman Index, calculated as the sum of the squared market shares of firms within the same industry, with a higher value indicating greater industry concentration and less competition.

Table 7 shows that the coefficient on the triple interaction term *Treated* × *Post* × *Cash* is significantly negative, suggesting that firms with stronger liquidity are less exposed to the crash risk induced by the AI policy. In contrast, the coefficient on *Treated* × *Post* × *HHI* is significantly positive, indicating that the policy's impact on crash risk is more pronounced in more concentrated industries. Overall, these findings underscore that resource constraints—both financial and competitive—serve as important moderators shaping the economic consequences of the AI pilot policy.

**Table 8**  
Economic consequences of AI investment.

|                               | (1)                          | (2)                          | (3)                     |
|-------------------------------|------------------------------|------------------------------|-------------------------|
|                               | <i>ADM_Fee<sub>t+1</sub></i> | <i>Sup_Fin<sub>t+1</sub></i> | <i>RD<sub>t+1</sub></i> |
| <i>Treat</i> × <i>Post</i>    | −0.006***<br>(−3.00)         | −0.019***<br>(−5.64)         | −0.001**<br>(−2.12)     |
| <i>Size<sub>t</sub></i>       | −0.008***<br>(−10.80)        | −0.006***<br>(−3.80)         | 0.002***<br>(6.92)      |
| <i>Lev<sub>t</sub></i>        | −0.048***<br>(−8.95)         | 0.249***<br>(29.09)          | −0.003**<br>(−2.42)     |
| <i>ROA<sub>t</sub></i>        | −0.134***<br>(−9.69)         | 0.055***<br>(3.20)           | −0.017***<br>(−5.28)    |
| <i>PAF<sub>t</sub></i>        | 0.014**<br>(2.26)            | −0.119***<br>(−11.62)        | −0.002<br>(−0.95)       |
| <i>SOE<sub>t</sub></i>        | −0.003*<br>(−1.85)           | 0.017***<br>(4.68)           | −0.001<br>(−0.88)       |
| <i>Growth<sub>t</sub></i>     | −0.008***<br>(−4.72)         | −0.001<br>(−0.22)            | −0.001<br>(−0.82)       |
| <i>Top1<sub>t</sub></i>       | −0.022***<br>(−4.14)         | 0.037***<br>(3.65)           | −0.005***<br>(−2.81)    |
| <i>Mshare<sub>t</sub></i>     | −0.018**<br>(−2.11)          | 0.027**<br>(1.99)            | 0.007**<br>(2.19)       |
| <i>Duality<sub>t</sub></i>    | 0.001<br>(0.64)              | −0.005*<br>(−1.72)           | −0.001<br>(−0.51)       |
| <i>Separation<sub>t</sub></i> | 0.003<br>(0.34)              | 0.044**<br>(2.36)            | 0.001<br>(0.14)         |
| <i>Intensity<sub>t</sub></i>  | 0.008***<br>(15.49)          | −0.006***<br>(−9.71)         | 0.001***<br>(3.33)      |
| <i>BM<sub>t</sub></i>         | −0.025***<br>(−6.36)         | 0.015**<br>(2.26)            | −0.005***<br>(−4.53)    |
| <i>RET<sub>t</sub></i>        | −0.425***<br>(−6.53)         | 0.275***<br>(2.82)           | −0.052***<br>(−3.31)    |
| <i>Sigma<sub>t</sub></i>      | 0.075**<br>(2.50)            | −0.040<br>(−0.81)            | 0.014*<br>(1.94)        |
| <i>Turnover<sub>t</sub></i>   | 0.001***<br>(4.67)           | −0.001<br>(−0.76)            | 0.001<br>(1.44)         |
| _cons                         | 0.285***<br>(18.20)          | 0.189***<br>(5.98)           | −0.024***<br>(−5.67)    |
| <i>Industry FE</i>            | Y                            | Y                            | Y                       |
| <i>Year FE</i>                | Y                            | Y                            | Y                       |
| <i>City FE</i>                | Y                            | Y                            | Y                       |
| <i>N</i>                      | 21,806                       | 21,806                       | 21,806                  |
| adj. <i>R</i> <sup>2</sup>    | 0.47                         | 0.50                         | 0.19                    |

This table reports the results of Economic consequences of AI pilot zone policy. we examine its impact on firms' administrative expenses (*ADM\_Fee*), supply chain finance (*Sup\_Fin*), and R&D investment (*RD*). Robust t-values clustered at the firm level are reported in parentheses. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

### 3.3. Additional exploratory analysis (unregistered)

To further explore the broader economic consequences of the AI pilot policy, we investigate its impact on firms' cost structures, financing strategies, and innovation activities. Specifically, we examine changes in administrative expenses, reliance on supply chain finance, and R&D investment following the policy's implementation. Administrative expenses (*ADM\_Fee*) are measured as the ratio of administrative expenses to operating revenue, supply chain finance (*Sup\_Fin*) is proxied by the sum of notes payable, accounts payable, and advances from customers scaled by total assets, and R&D investment (*RD*) is calculated as the ratio of R&D expenditures to operating revenue.

The regression results are presented in Table 8. The coefficients of *Treat*×*Post* are negative across all columns, indicating that following the implementation of the AI pilot policy, firms experience a significant reduction in administrative expenses and a decreased reliance on supply chain finance. One possible explanation is that the inclusion in the pilot zones enables firms to improve firms' internal management efficiency and access alternative financing channels—such as policy loans, government funds, or tax incentives—which in turn alleviates the need for supply chain-based financing. However, we also find that R&D investment is significantly reduced, suggesting a potential resource crowding-out effect, where firms reallocate resources from long-term innovation activities toward AI-related expenditures or other short-term strategic adjustments. These findings underscore the nuanced economic consequences of the AI policy, which may simultaneously enhance financial flexibility while dampening firms' incentives to engage in innovation-driven growth.

#### 4. Discussion

The baseline results provide evidence that firms subject to the AI pilot zone policy experience a higher stock price crash risk compared with those unaffected, providing support for H1b. The robustness of our findings is confirmed through a series of tests, including parallel trend analysis, placebo tests, and PSM-DID.

The mechanism analysis indicates that AI investment increases stock price crash risk by reducing information transparency and amplifying managerial optimism. Additional tests show that firm value increases following the AI pilot zone policy, reflecting an environment of excessive optimism in which markets initially overreact to the perceived benefits of AI adoption, thereby inflating valuations. Cross-sectional analyses further reveal that the positive impact of the AI policy on stock price crash risk is more pronounced among firms with lower information transparency and greater resource constraints.

#### 5. Conclusion

In this study, we employ a DID model based on the “New-generation Artificial Intelligence Pilot Zone” policy launched by the Chinese government, to investigate the impact of AI investment on stock price crash risk and firm value. Our findings indicate that AI investment significantly increases stock price crash risk by reducing information transparency and amplifying managerial optimism. Additionally, we find that firm value rises following the implementation of the AI pilot policy, reflecting an environment of excessive optimism in which markets initially overreact to the perceived benefits of AI adoption. Cross-sectional analyses further reveal that the positive effect of AI investment on crash risk is more pronounced among firms with lower information transparency and tighter resource constraints. Our main findings remain robust across a series of checks.

Our findings highlight the risk-enhancing effect of AI investment and underscore the importance of prudent policy design when promoting AI-driven development. In this regard, policymakers should exercise caution when promoting large-scale AI initiatives, as the heightened information opacity and managerial overconfidence associated with AI adoption may elevate stock price crash risk and threaten financial stability. In addition, firms should strengthen internal governance and improve disclosure quality to reduce the accumulation of hidden losses and mitigate extreme downside events. Furthermore, our evidence on the coexistence of short-term valuation gains and long-term crash risk provides important implications for investors and regulators. Moreover, the coexistence of short-term valuation gains and long-term crash risk points to an important trade-off for investors and regulators. While AI-related policies may temporarily boost firm value due to market optimism, such gains can mask underlying vulnerabilities. Incorporating this trade-off into investment decisions and regulatory oversight is essential to maintain long-term financial stability.

#### CRediT authorship contribution statement

**Xiao Bai:** Writing – original draft, Software, Methodology, Investigation, Data curation, Conceptualization. **Wenyao Zhao:** Visualization, Validation, Software, Methodology, Investigation. **Meng Liu:** Writing – review & editing, Visualization, Methodology, Funding acquisition.

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#### Declaration of competing interest

None.

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## Appendix A. Appendix

### A.1. Variable definition

The following table offers comprehensive definitions of dependent, independent, and control variables of the paper.

| Variables            |                                                                                                                                                                | Measurement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Dependent variables  | $NCSKEW_{i,t+1}$<br>$DUVOL_{i,t+1}$                                                                                                                            | the negative conditional skewness of firm-specific weekly returns,<br>the down-to-up volatility ratio                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Independent variable | $Treat_{i,t}$<br>$Post_{i,t}$<br>$Size_{i,t}$<br>$Lev_{i,t}$<br>$ROA_{i,t}$<br>$PFA_{i,t}$<br>$SOE_{i,t}$<br>$Growth_{i,t}$                                    | an indicator for the treatment and equals 1(0) if a firm is in the treatment (control) group<br>the time indicator which equals 1(0) if a firm is in the post-event(pre-event) period<br>the natural logarithm of total asset<br>total liabilities divided by total assets<br>net profit divided by total assets<br>fixed assets divided by total assets<br>equals 1 if the firm is state-owned, and 0 otherwise.<br>the growth rate of operating revenue                                                                                                                       |
| Controls             | $Top1_{i,t}$<br>$Mshare_{i,t}$<br>$Duality_{i,t}$<br>$Separation_{i,t}$<br>$Intensity_{i,t}$<br>$BM_{i,t}$<br>$RET_{i,t}$<br>$Sigma_{i,t}$<br>$Turnover_{i,t}$ | the proportion of the largest shareholder's holding<br>percentage of managers' shareholding<br>equals 1 if the roles of chairman and general manager are held by the same person, and 0 otherwise.<br>the degree of separation between ownership and control<br>total assets divided by operating revenue<br>book value per share divided by market value per share<br>the mean of firm-specific weekly returns over the fiscal year period<br>the standard deviation of firm-specific weekly returns over the fiscal year period<br>share turnover over the fiscal year period |

### A.2. Results of parallel trend test

The following table presents the results of parallel trend test based on Eq. (2) in the robustness test section (Section 3.2.2.1).

|                        | (1)                  | (2)                  |
|------------------------|----------------------|----------------------|
|                        | $NCSKEW_{t+1}$       | $DUVOL_{t+1}$        |
| $Treat \times Pre\_4$  | 0.012<br>(0.34)      | 0.011<br>(0.48)      |
| $Treat \times Pre\_3$  | 0.010<br>(0.29)      | 0.009<br>(0.41)      |
| $Treat \times Pre\_2$  | -0.013<br>(-0.40)    | -0.019<br>(-0.95)    |
| $Treat \times Pre\_1$  | 0.031<br>(1.04)      | 0.020<br>(1.13)      |
| $Treat \times Post\_1$ | 0.063**<br>(2.21)    | 0.051***<br>(3.01)   |
| $Treat \times Post\_2$ | 0.077***<br>(2.81)   | 0.048***<br>(2.94)   |
| $Treat \times Post\_3$ | 0.089***<br>(2.83)   | 0.038**<br>(2.04)    |
| $NCSKEW_t$             | 0.042***<br>(6.00)   |                      |
| $DUVOL_t$              |                      | 0.056***<br>(8.23)   |
| $Size_t$               | 0.072***<br>(12.59)  | 0.033***<br>(9.17)   |
| $Lev_t$                | -0.056<br>(-1.61)    | -0.037*<br>(-1.74)   |
| $ROA_t$                | -0.232**<br>(-2.35)  | -0.150**<br>(-2.43)  |
| $PAF_t$                | 0.003<br>(0.08)      | -0.018<br>(-0.74)    |
| $SOE_t$                | -0.057***<br>(-4.45) | -0.038***<br>(-4.99) |
| $Growth_t$             | -0.004<br>(-0.27)    | -0.003<br>(-0.26)    |

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|                     | (1)                   | (2)                   |
|---------------------|-----------------------|-----------------------|
|                     | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         |
| $Top1_t$            | −0.031<br>(−0.79)     | 0.001<br>(0.04)       |
| $Mshare_t$          | 0.174***<br>(2.71)    | 0.080**<br>(1.99)     |
| $Duality_t$         | 0.016<br>(1.19)       | 0.009<br>(1.10)       |
| $Separation_t$      | −0.053<br>(−0.76)     | −0.003<br>(−0.07)     |
| $Intensity_t$       | 0.001<br>(0.17)       | 0.001<br>(0.73)       |
| $BM_t$              | −0.440***<br>(−15.06) | −0.230***<br>(−12.62) |
| $RET_t$             | 12.000***<br>(14.49)  | 8.746***<br>(15.83)   |
| $Sigma_t$           | −0.905***<br>(−2.72)  | −1.134***<br>(−5.39)  |
| $Turnover_t$        | −0.001***<br>(−5.93)  | −0.001***<br>(−5.87)  |
| _cons               | −1.668***<br>(−13.81) | −0.790***<br>(−10.52) |
| Industry FE         | Y                     | Y                     |
| Year FE             | Y                     | Y                     |
| City FE             | Y                     | Y                     |
| N                   | 25,746                | 25,746                |
| adj. R <sup>2</sup> | 0.06                  | 0.06                  |

## A.3. Results of PSM-DID

The following table details the results of PSM-DID in the robustness test section (Section 3.2.2.3).

|                       | Nearest neighbor matching |                      | Radius matching      |                      | Kernel matching      |                      |
|-----------------------|---------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                       | (1)                       | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|                       | $NCSKEW_{t+1}$            | $DUVOL_{t+1}$        | $NCSKEW_{t+1}$       | $DUVOL_{t+1}$        | $NCSKEW_{t+1}$       | $DUVOL_{t+1}$        |
| $Treated \times Post$ | 0.062***<br>(3.00)        | 0.045***<br>(3.39)   | 0.050***<br>(2.87)   | 0.030***<br>(2.61)   | 0.063***<br>(3.14)   | 0.047***<br>(3.66)   |
| $NCSKEW_t$            | 0.030***<br>(3.70)        |                      | 0.039***<br>(5.90)   |                      | 0.036***<br>(4.43)   |                      |
| $DUVOL_t$             |                           | 0.032***<br>(4.06)   |                      | 0.042***<br>(6.17)   |                      | 0.039***<br>(4.98)   |
| $Size_t$              | 0.077***<br>(11.83)       | 0.037***<br>(8.82)   | 0.075***<br>(13.76)  | 0.036***<br>(9.57)   | 0.071***<br>(10.90)  | 0.034***<br>(8.28)   |
| $Lev_t$               | −0.072*<br>(−1.82)        | −0.046*<br>(−1.85)   | −0.042<br>(−1.31)    | −0.038*<br>(−1.73)   | −0.035<br>(−0.88)    | −0.026<br>(−1.04)    |
| $ROA_t$               | −0.259**<br>(−2.30)       | −0.198***<br>(−2.75) | −0.218**<br>(−2.46)  | −0.174***<br>(−2.73) | −0.205*<br>(−1.81)   | −0.170**<br>(−2.37)  |
| $PAF_t$               | 0.007<br>(0.15)           | −0.018<br>(−0.61)    | 0.040<br>(0.95)      | −0.023<br>(−0.90)    | 0.070<br>(1.47)      | 0.017<br>(0.54)      |
| $SOE_t$               | −0.060***<br>(−4.09)      | −0.041***<br>(−4.36) | −0.056***<br>(−4.59) | −0.043***<br>(−5.24) | −0.054***<br>(−3.56) | −0.042***<br>(−4.37) |
| $Growth_t$            | 0.006<br>(0.31)           | 0.009<br>(0.78)      | 0.009<br>(0.59)      | 0.001<br>(0.03)      | 0.022<br>(1.14)      | 0.016<br>(1.31)      |
| $Top1_t$              | −0.054<br>(−1.19)         | −0.004<br>(−0.14)    | −0.062*<br>(−1.76)   | 0.005<br>(0.20)      | −0.035<br>(−0.79)    | 0.006<br>(0.21)      |
| $Mshare_t$            | 0.156**<br>(2.16)         | 0.086*<br>(1.85)     | 0.199***<br>(3.17)   | 0.093**<br>(2.23)    | 0.199***<br>(2.69)   | 0.124**<br>(2.56)    |
| $Duality_t$           | 0.013<br>(0.85)           | 0.011<br>(1.08)      | 0.013<br>(0.97)      | 0.009<br>(1.02)      | 0.020<br>(1.34)      | 0.011<br>(1.12)      |
| $Separation_t$        | −0.112<br>(−1.39)         | −0.056<br>(−1.06)    | −0.065<br>(−0.96)    | −0.017<br>(−0.37)    | −0.080<br>(−0.97)    | −0.028<br>(−0.52)    |
| $Intensity_t$         | 0.001<br>(0.15)           | 0.001<br>(0.77)      | 0.001<br>(0.76)      | 0.001<br>(0.63)      | 0.001<br>(0.19)      | 0.001<br>(0.85)      |
| $BM_t$                | −0.434***                 | −0.247***            | −0.468***            | −0.247***            | −0.458***            | −0.256***            |

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|                                     | Nearest neighbor matching           |                                    | Radius matching                     |                                    | Kernel matching                     |                                    |
|-------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|------------------------------------|-------------------------------------|------------------------------------|
|                                     | (1)                                 | (2)                                | (3)                                 | (4)                                | (5)                                 | (6)                                |
|                                     | <i>NCSKEW</i> <sub><i>t</i>+1</sub> | <i>DUVOL</i> <sub><i>t</i>+1</sub> | <i>NCSKEW</i> <sub><i>t</i>+1</sub> | <i>DUVOL</i> <sub><i>t</i>+1</sub> | <i>NCSKEW</i> <sub><i>t</i>+1</sub> | <i>DUVOL</i> <sub><i>t</i>+1</sub> |
| <i>RET</i> <sub><i>t</i></sub>      | (−12.87)<br>11.260***<br>(11.71)    | (−11.32)<br>7.736***<br>(12.02)    | (−16.39)<br>11.94***<br>(13.93)     | (−13.06)<br>8.482***<br>(15.26)    | (−13.64)<br>12.11***<br>(12.91)     | (−11.99)<br>8.616***<br>(13.81)    |
| <i>Sigma</i> <sub><i>t</i></sub>    | −0.733*<br>(−1.90)                  | −0.840***<br>(−3.40)               | −0.804**<br>(−2.45)                 | −1.101***<br>(−5.18)               | −0.841**<br>(−2.18)                 | −1.070***<br>(−4.34)               |
| <i>Turnover</i> <sub><i>t</i></sub> | −0.001***<br>(−4.24)                | −0.001***<br>(−4.49)               | −0.001***<br>(−5.88)                | −0.001***<br>(−5.96)               | −0.001***<br>(−5.46)                | −0.001***<br>(−5.61)               |
| _cons                               | −1.777***<br>(−13.07)               | −0.890***<br>(−10.17)              | −1.731***<br>(−15.27)               | −0.841***<br>(−10.83)              | −1.673***<br>(−12.33)               | −0.837***<br>(−9.68)               |
| <i>Industry FE</i>                  | Y                                   | Y                                  | Y                                   | Y                                  | Y                                   | Y                                  |
| <i>Year FE</i>                      | Y                                   | Y                                  | Y                                   | Y                                  | Y                                   | Y                                  |
| <i>City FE</i>                      | Y                                   | Y                                  | Y                                   | Y                                  | Y                                   | Y                                  |
| <i>N</i>                            | 18,900                              | 18,900                             | 25,732                              | 25,731                             | 25,732                              | 25,732                             |
| adj. <i>R</i> <sup>2</sup>          | 0.06                                | 0.06                               | 0.07                                | 0.06                               | 0.06                                | 0.07                               |

#### A.4. Results based on alternative measures of stock price crash risk

The following table details the results of alternative measures of stock price crash risk in the robustness test section (Section 3.2.2.4).

|                                       | (1)                                 | (2)                                 |
|---------------------------------------|-------------------------------------|-------------------------------------|
|                                       | <i>CRASH1</i> <sub><i>t</i>+1</sub> | <i>CRASH2</i> <sub><i>t</i>+1</sub> |
| <i>Treated</i> × <i>Post</i>          | 0.018**<br>(2.17)                   | 0.015*<br>(1.94)                    |
| <i>Size</i> <sub><i>t</i></sub>       | −0.015***<br>(−6.19)                | −0.012***<br>(−5.14)                |
| <i>Lev</i> <sub><i>t</i></sub>        | 0.020<br>(1.33)                     | 0.018<br>(1.19)                     |
| <i>ROA</i> <sub><i>t</i></sub>        | −0.088*<br>(−1.94)                  | −0.090**<br>(−2.00)                 |
| <i>PAF</i> <sub><i>t</i></sub>        | −0.037**<br>(−2.18)                 | −0.021<br>(−1.24)                   |
| <i>SOE</i> <sub><i>t</i></sub>        | −0.006<br>(−1.16)                   | −0.005<br>(−1.02)                   |
| <i>Growth</i> <sub><i>t</i></sub>     | −0.005<br>(−0.72)                   | −0.003<br>(−0.37)                   |
| <i>Top1</i> <sub><i>t</i></sub>       | 0.005<br>(0.30)                     | 0.006<br>(0.37)                     |
| <i>Mshare</i> <sub><i>t</i></sub>     | 0.005<br>(0.18)                     | −0.004<br>(−0.13)                   |
| <i>Duality</i> <sub><i>t</i></sub>    | 0.005<br>(0.91)                     | 0.010<br>(1.63)                     |
| <i>Separation</i> <sub><i>t</i></sub> | −0.056**<br>(−1.98)                 | −0.057**<br>(−2.01)                 |
| <i>Intensity</i> <sub><i>t</i></sub>  | −0.001<br>(−0.06)                   | −0.001<br>(−0.36)                   |
| <i>BM</i> <sub><i>t</i></sub>         | 0.002<br>(0.18)                     | −0.012<br>(−0.88)                   |
| <i>RET</i> <sub><i>t</i></sub>        | 1.065***<br>(2.99)                  | 0.894**<br>(2.51)                   |
| <i>Sigma</i> <sub><i>t</i></sub>      | −0.471***<br>(−3.20)                | −0.502***<br>(−3.51)                |
| <i>Turnover</i> <sub><i>t</i></sub>   | −0.001***<br>(−3.70)                | −0.001***<br>(−4.14)                |
| _cons                                 | 0.447***<br>(8.92)                  | 0.399***<br>(7.93)                  |
| <i>Industry FE</i>                    | Y                                   | Y                                   |
| <i>Year FE</i>                        | Y                                   | Y                                   |

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|                     | (1)                          | (2)                          |
|---------------------|------------------------------|------------------------------|
|                     | <i>CRASH1</i> <sub>t+1</sub> | <i>CRASH2</i> <sub>t+1</sub> |
| City FE             | Y                            | Y                            |
| N                   | 21,814                       | 21,814                       |
| adj. R <sup>2</sup> | 0.02                         | 0.02                         |

Table 5 reports the results using alternative measures of stock price crash risk. Columns (1) and (2) show the results where *CRASH1* and *CRASH2* are the dependent variable, respectively. The robust t-values are clustered at the firm level and presented in parentheses. \*, \*\*, and \*\*\* indicate statistical significance at the 10 %, 5 %, and 1 % levels, respectively.

#### A.5. Results based on alternative measures of AI investment

The following table details the results of alternative measures of AI investment in the robustness test section (Section 3.2.2.5).

|                                         | (1)                          | (2)                         | (3)                          | (4)                         |
|-----------------------------------------|------------------------------|-----------------------------|------------------------------|-----------------------------|
|                                         | <i>NCSKEW</i> <sub>t+1</sub> | <i>DUVOL</i> <sub>t+1</sub> | <i>NCSKEW</i> <sub>t+1</sub> | <i>DUVOL</i> <sub>t+1</sub> |
| <i>AI</i> <sub>level</sub> <sub>t</sub> | 0.362**<br>(2.42)            | 0.157**<br>(2.29)           |                              |                             |
| <i>lnAI</i> <sub>t</sub>                |                              |                             | 0.002**<br>(2.56)            | 0.002***<br>(2.87)          |
| <i>NCSKEW</i> <sub>t</sub>              | 0.042***<br>(5.98)           |                             | 0.041***<br>(5.40)           |                             |
| <i>DUVOL</i> <sub>t</sub>               |                              | 0.042***<br>(6.20)          |                              | 0.038***<br>(5.20)          |
| <i>Size</i> <sub>t</sub>                | 0.072***<br>(12.59)          | 0.036***<br>(9.58)          | 0.072***<br>(10.37)          | 0.034***<br>(7.47)          |
| <i>Lev</i> <sub>t</sub>                 | −0.055<br>(−1.60)            | −0.038*<br>(−1.70)          | −0.089**<br>(−2.42)          | −0.054**<br>(−2.34)         |
| <i>ROA</i> <sub>t</sub>                 | −0.239**<br>(−2.41)          | −0.176***<br>(−2.76)        | −0.182*<br>(−1.72)           | −0.139**<br>(−2.04)         |
| <i>PAF</i> <sub>t</sub>                 | −0.001<br>(−0.01)            | −0.025<br>(−0.97)           | −0.008<br>(−0.17)            | −0.036<br>(−1.27)           |
| <i>SOE</i> <sub>t</sub>                 | −0.058***<br>(−4.50)         | −0.044***<br>(−5.30)        | −0.059***<br>(−4.29)         | −0.042***<br>(−4.75)        |
| <i>Growth</i> <sub>t</sub>              | −0.005<br>(−0.28)            | 0.001<br>(0.04)             | −0.009<br>(−0.54)            | −0.001<br>(−0.03)           |
| <i>Top1</i> <sub>t</sub>                | −0.031<br>(−0.80)            | 0.005<br>(0.20)             | −0.029<br>(−0.70)            | 0.017<br>(0.66)             |
| <i>Mshare</i> <sub>t</sub>              | 0.176***<br>(2.73)           | 0.094**<br>(2.26)           | 0.145**<br>(2.16)            | 0.071*<br>(1.65)            |
| <i>Duality</i> <sub>t</sub>             | 0.016<br>(1.20)              | 0.009<br>(1.03)             | 0.010<br>(0.69)              | 0.006<br>(0.69)             |
| <i>Separation</i> <sub>t</sub>          | −0.052<br>(−0.75)            | −0.016<br>(−0.35)           | 0.008<br>(0.11)              | 0.009<br>(0.18)             |
| <i>Intensity</i> <sub>t</sub>           | 0.001<br>(0.23)              | 0.001<br>(0.67)             | 0.001<br>(0.15)              | 0.001<br>(0.80)             |
| <i>BM</i> <sub>t</sub>                  | −0.440***<br>(−15.03)        | −0.247***<br>(−13.05)       | −0.432***<br>(−13.78)        | −0.240***<br>(−11.77)       |
| <i>RET</i> <sub>t</sub>                 | 11.990***<br>(14.48)         | 8.477***<br>(15.25)         | 12.120***<br>(13.65)         | 8.648***<br>(14.43)         |
| <i>Sigma</i> <sub>t</sub>               | −0.932***<br>(−2.81)         | −1.110***<br>(−5.22)        | −0.815**<br>(−2.30)          | −1.063***<br>(−4.68)        |
| <i>Turnover</i> <sub>t</sub>            | −0.001***<br>(−5.92)         | −0.001***<br>(−5.93)        | −0.086***<br>(−5.46)         | −0.050***<br>(−5.14)        |
| _cons                                   | −1.657***<br>(−13.75)        | −0.835***<br>(−10.76)       | −1.692***<br>(−13.00)        | −0.832***<br>(−9.86)        |
| Industry FE                             | Y                            | Y                           | Y                            | Y                           |
| Year FE                                 | Y                            | Y                           | Y                            | Y                           |
| City FE                                 | Y                            | Y                           | Y                            | Y                           |
| N                                       | 25,746                       | 25,746                      | 25,746                       | 25,746                      |
| adj. R <sup>2</sup>                     | 0.06                         | 0.06                        | 0.06                         | 0.06                        |

## A.6. Results with firm-fixed effects included

The following table details the results with firm-fixed effects included in the robustness test section (Section 3.2.2.6).

|                               | (1)                   | (2)                   |
|-------------------------------|-----------------------|-----------------------|
|                               | $NCSKEW_{t+1}$        | $DUVOL_{t+1}$         |
| <i>Treated</i> × <i>Post</i>  | 0.046**<br>(2.09)     | 0.028**<br>(2.04)     |
| <i>NCSKEW<sub>t</sub></i>     | −0.071***<br>(−8.98)  |                       |
| <i>DUVOL<sub>t</sub></i>      |                       | −0.072***<br>(−10.44) |
| <i>Size<sub>t</sub></i>       | 0.147***<br>(10.08)   | 0.087***<br>(9.09)    |
| <i>Lev<sub>t</sub></i>        | −0.271***<br>(−4.16)  | −0.168***<br>(−4.62)  |
| <i>ROA<sub>t</sub></i>        | −0.622***<br>(−4.23)  | −0.432***<br>(−5.66)  |
| <i>PAF<sub>t</sub></i>        | −0.202***<br>(−2.66)  | −0.132***<br>(−2.76)  |
| <i>SOE<sub>t-1</sub></i>      | −0.091**<br>(−2.48)   | −0.071***<br>(−2.92)  |
| <i>Growth<sub>t</sub></i>     | 0.004<br>(0.23)       | 0.006<br>(0.55)       |
| <i>Top1<sub>t</sub></i>       | 0.226**<br>(2.05)     | 0.128**<br>(2.00)     |
| <i>Mshare<sub>t</sub></i>     | 0.109<br>(0.98)       | −0.009<br>(−0.11)     |
| <i>Duality<sub>t</sub></i>    | 0.041**<br>(2.15)     | 0.021<br>(1.61)       |
| <i>Separation<sub>t</sub></i> | −0.152<br>(−1.15)     | −0.074<br>(−0.80)     |
| <i>Intensity<sub>t</sub></i>  | 0.004<br>(1.38)       | 0.003*<br>(1.70)      |
| <i>BM<sub>t</sub></i>         | −0.594***<br>(−11.81) | −0.379***<br>(−12.91) |
| <i>RET<sub>t</sub></i>        | 9.406***<br>(11.83)   | 6.611***<br>(11.37)   |
| <i>Sigma<sub>t</sub></i>      | −0.317<br>(−0.83)     | −0.622**<br>(−2.57)   |
| <i>Turnover<sub>t</sub></i>   | −0.001***<br>(−4.90)  | −0.001***<br>(−5.66)  |
| _cons                         | −3.255***<br>(−10.47) | −1.922***<br>(−9.13)  |
| <i>Firm FE</i>                | Y                     | Y                     |
| <i>Year FE</i>                | Y                     | Y                     |
| <i>City FE</i>                | Y                     | Y                     |
| <i>N</i>                      | 25,746                | 25,746                |
| adj. <i>R</i> <sup>2</sup>    | 0.09                  | 0.09                  |

## A.7. Results using city-level clustering

The table below summarizes the results of city-level clustering used in the robustness checks (Section 3.2.2.7).

|                              | (1)                 | (2)                 |
|------------------------------|---------------------|---------------------|
|                              | $NCSKEW_{t+1}$      | $DUVOL_{t+1}$       |
| <i>Treated</i> × <i>Post</i> | 0.043**<br>(1.99)   | 0.030**<br>(2.39)   |
| <i>NCSKEW<sub>t</sub></i>    | 0.042***<br>(4.73)  |                     |
| <i>DUVOL<sub>t</sub></i>     |                     | 0.042***<br>(4.84)  |
| <i>Size<sub>t</sub></i>      | 0.072***<br>(12.87) | 0.036***<br>(10.08) |

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|                                       | (1)                                 | (2)                                |
|---------------------------------------|-------------------------------------|------------------------------------|
|                                       | <i>NCSKEW</i> <sub><i>t</i>+1</sub> | <i>DUVOL</i> <sub><i>t</i>+1</sub> |
| <i>Lev</i> <sub><i>t</i></sub>        | −0.056<br>(−1.36)                   | −0.038<br>(−1.41)                  |
| <i>ROA</i> <sub><i>t</i></sub>        | −0.237*<br>(−1.92)                  | −0.174**<br>(−2.26)                |
| <i>PAF</i> <sub><i>t</i></sub>        | 0.003<br>(0.06)                     | −0.023<br>(−0.95)                  |
| <i>SOE</i> <sub><i>t</i></sub>        | −0.057***<br>(−4.04)                | −0.043***<br>(−4.75)               |
| <i>Growth</i> <sub><i>t</i></sub>     | −0.005<br>(−0.31)                   | 0.001<br>(0.03)                    |
| <i>Top1</i> <sub><i>t</i></sub>       | −0.031<br>(−0.59)                   | 0.005<br>(0.15)                    |
| <i>Mshare</i> <sub><i>t</i></sub>     | 0.174***<br>(2.71)                  | 0.093**<br>(2.33)                  |
| <i>Duality</i> <sub><i>t</i></sub>    | 0.016<br>(1.17)                     | 0.009<br>(0.95)                    |
| <i>Separation</i> <sub><i>t</i></sub> | −0.053<br>(−0.81)                   | −0.017<br>(−0.40)                  |
| <i>Intensity</i> <sub><i>t</i></sub>  | 0.001<br>(0.16)                     | 0.001<br>(0.58)                    |
| <i>BM</i> <sub><i>t</i></sub>         | −0.440***<br>(−13.12)               | −0.247***<br>(−12.10)              |
| <i>RET</i> <sub><i>t</i></sub>        | 12.000***<br>(15.91)                | 8.482***<br>(17.51)                |
| <i>Sigma</i> <sub><i>t</i></sub>      | −0.918***<br>(−2.89)                | −1.101***<br>(−6.22)               |
| <i>Turnover</i> <sub><i>t</i></sub>   | −0.001***<br>(−5.13)                | −0.001***<br>(−4.84)               |
| _cons                                 | −1.665***<br>(−14.49)               | −0.841***<br>(−11.79)              |
| <i>Industry FE</i>                    | Y                                   | Y                                  |
| <i>Year FE</i>                        | Y                                   | Y                                  |
| <i>City FE</i>                        | Y                                   | Y                                  |
| <i>N</i>                              | 25,746                              | 25,746                             |
| adj. <i>R</i> <sup>2</sup>            | 0.06                                | 0.06                               |

## Appendix B. Heterogeneity analysis: high-tech vs. non-high-tech firms

We further examine whether the impact of AI investment on stock price crash risk varies by firms' technological attributes. The classification of high-tech enterprises follows the *Measures for the Administration of the Recognition of High-Tech Enterprises* (revised in 2016 by the Ministry of Science and Technology, Ministry of Finance, and State Administration of Taxation of China).

As reported in following table, the results show that AI investment does not significantly affect crash risk for high-tech firms, but it significantly increases crash risk for non-high-tech firms. This suggests that firms with weaker technological capabilities may face greater challenges in managing AI-related information opacity, leading to higher downside risk.

|                                   | High-Tech Firms                     |                                    | Non-High-Tech Firms                 |                                    |
|-----------------------------------|-------------------------------------|------------------------------------|-------------------------------------|------------------------------------|
|                                   | (1)                                 | (2)                                | (3)                                 | (4)                                |
|                                   | <i>NCSKEW</i> <sub><i>t</i>+1</sub> | <i>DUVOL</i> <sub><i>t</i>+1</sub> | <i>NCSKEW</i> <sub><i>t</i>+1</sub> | <i>DUVOL</i> <sub><i>t</i>+1</sub> |
| <i>Treat</i> × <i>Post</i>        | −0.092<br>(−0.92)                   | −0.026<br>(−0.41)                  | 0.043**<br>(2.34)                   | 0.029**<br>(2.43)                  |
| <i>NCSKEW</i> <sub><i>t</i></sub> | −0.079**<br>(−2.19)                 |                                    | 0.042***<br>(5.80)                  |                                    |
| <i>DUVOL</i> <sub><i>t</i></sub>  |                                     | −0.032<br>(−0.90)                  |                                     | 0.040***<br>(5.78)                 |
| <i>Size</i> <sub><i>t</i></sub>   | 0.132***<br>(3.99)                  | 0.070***<br>(3.61)                 | 0.072***<br>(12.25)                 | 0.035***<br>(9.24)                 |
| <i>Lev</i> <sub><i>t</i></sub>    | −0.104<br>(−0.43)                   | −0.114<br>(−0.81)                  | −0.054<br>(−1.54)                   | −0.035<br>(−1.56)                  |
| <i>ROA</i> <sub><i>t</i></sub>    | 0.476<br>(0.45)                     | 0.082<br>(0.14)                    | −0.242**<br>(−2.43)                 | −0.175***<br>(−2.72)               |
| <i>PAF</i> <sub><i>t</i></sub>    | 0.086                               | −0.096                             | 0.001                               | −0.023                             |

(continued on next page)

(continued)

|                     | High-Tech Firms             |                                | Non-High-Tech Firms            |                                 |
|---------------------|-----------------------------|--------------------------------|--------------------------------|---------------------------------|
|                     | (1)                         | (2)                            | (3)                            | (4)                             |
|                     | $NCSKEW_{t+1}$              | $DUVOL_{t+1}$                  | $NCSKEW_{t+1}$                 | $DUVOL_{t+1}$                   |
| SOE                 | (0.46)<br>−0.119<br>(−1.54) | (−0.82)<br>−0.116**<br>(−2.40) | (0.02)<br>−0.054***<br>(−4.11) | (−0.82)<br>−0.041***<br>(−4.87) |
| $Growth_t$          | −0.055<br>(−0.69)           | −0.087*<br>(−1.80)             | −0.003<br>(−0.18)              | 0.003<br>(0.30)                 |
| $Top1_t$            | 0.003<br>(0.01)             | 0.107<br>(0.69)                | −0.029<br>(−0.73)              | 0.002<br>(0.06)                 |
| $Mshare_t$          | −0.215<br>(−0.31)           | 0.132<br>(0.29)                | 0.180***<br>(2.80)             | 0.094**<br>(2.26)               |
| $Duality_t$         | 0.177<br>(1.65)             | 0.048<br>(0.66)                | 0.013<br>(0.93)                | 0.008<br>(0.90)                 |
| $Separation_t$      | 0.120<br>(0.38)             | −0.102<br>(−0.52)              | −0.059<br>(−0.83)              | −0.012<br>(−0.26)               |
| $Intensity_t$       | −0.005<br>(−0.74)           | −0.005<br>(−1.54)              | 0.001<br>(0.65)                | 0.001<br>(1.06)                 |
| $BM_t$              | −0.784***<br>(−4.78)        | −0.388***<br>(−4.02)           | −0.440***<br>(−14.71)          | −0.248***<br>(−12.79)           |
| $RET_t$             | 11.230**<br>(2.01)          | 7.417**<br>(2.00)              | 11.790***<br>(14.05)           | 8.318***<br>(14.79)             |
| $Sigma_t$           | 0.064<br>(0.03)             | 0.402<br>(0.30)                | −0.952***<br>(−2.82)           | −1.126***<br>(−5.20)            |
| $Turnover_t$        | −0.001<br>(−0.25)           | 0.001<br>(0.26)                | −0.001***<br>(−5.92)           | −0.001***<br>(−5.99)            |
| _cons               | −2.912***<br>(−3.88)        | −1.531***<br>(−3.56)           | −1.662***<br>(−13.47)          | −0.834***<br>(−10.47)           |
| Industry FE         | Y                           | Y                              | Y                              | Y                               |
| Year FE             | Y                           | Y                              | Y                              | Y                               |
| City FE             | Y                           | Y                              | Y                              | Y                               |
| N                   | 1002                        | 1002                           | 24,727                         | 24,727                          |
| adj. R <sup>2</sup> | 0.21                        | 0.21                           | 0.06                           | 0.06                            |

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